Every Natural Strengthening Examined Is False: Type Mismatch in Lossless Reformulations of the Lonely Runner Conjecture

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Abstract

The Lonely Runner Conjecture (LRC_n) admits many *lossless* reformulations—finite, geometric, dynamical, or arithmetic rewrites exactly equivalent to it. This paper reports a structural fact that emerged from a program of attacking the conjecture through such reformulations: *every natural strengthening of a lossless reformulation that the program examined—every version that would open a new toolkit—is false.* The claim is an observational summary of four witnessed settings, not a meta-theorem over all conceivable strengthenings: a reformulation equivalent to LRC carries the full truth-content of LRC, and a strictly stronger version is false, with the falsity visible at small n through exact rational witnesses. We exhibit the type mismatch in four settings across two reformulation worlds. The finite pair-sum covering reformulation supports three: its proved inventory contains only closure, invariance, and exhaustiveness statements (no proved margin-type statement exists in it); its single margin-shaped instrument, a transfer-matrix model of the solution cascade, is closed by theorem *in its single-anchor scope* (the constrained transfer matrices are partial permutations of spectral radius exactly 0 or 1; the general branching strand-census operator is not covered); and its character-rigidity route has orbit-invariance laws falsified exactly, with marginal envelopes saturated precisely where the missing margin would be needed. The fourth setting is independent of the cascade: the Lonely Runner zonotope, whose covering radius satisfies a strictly stronger inequality than LRC that is false from $n = 5$ on, with $\rho = 16/47 > 1/3$ at $v = (1, 2, 3, 4, 5)$ (certified from both sides by branch and bound), strict witnesses $\rho \geq 17/47 > 5/14$ at $v = (1, 2, 3, 4, 5, 12)$, $\rho \geq 29/80$, $4/11$, $1793/5000$ (all $> 5/14$), and a proved weight identity among the quotient norm's pair forms that forces the denominators of the exact values (the recurring 47 is $3 \cdot 4 + 3 \cdot 5 + 4 \cdot 5$); its Ehrhart h^*-polynomials are real-rooted and log-concave through $n = 10$, with strictness decaying like $\Theta(1/n)$ and no correlation with covering margins at the achievable sample sizes. Alongside the negative result the paper states the program's one bounded positive theorem—the sum-only equality $M(V) = \max_{\{u,w\}} \tau(u, w)$, the rigid-zone translate-class classifications at $7, 11, 13, 17, 19, 37$, and the composite-side structure at p^2 and pq, with its hypothesis ledger—and a finite witness archive with a certification ladder: exact deepest-hole classification at $n = 3$, away-certified and branch-and-bound certificates at $n = 4$ ($\rho = 9/34$ at $v = (1, 2, 3, 16)$), exact refutations at $n = 5, 6$ (the sharpest two-sided), and certified deepest-hole equality at $n = 5$, $m = 10, 20, 25, 30$ ($\rho = \delta$ exactly). Every number is re-asserted by a persisted, independent, exact computation, and the certification method carries a negative control that correctly refuses a false statement. The paper is a negative-results paper and claims no progress on the conjecture itself: it proves no new cases of LRC, and the surviving equivalent forms are as hard as they ever were. Its contribution is the obstruction's anatomy—why four natural toolkits, each correct and each productive on its own terms, cannot cross the same gap.

Keywords: Lonely Runner Conjecture; lossless reformulation; type mismatch; covering radius; zonotope; Ehrhart h^*-polynomial; negative results.

1 Introduction

1.1 The conjecture and the reformulation habit

Let $n \geq 2$ and let $V = (v_1, \ldots, v_n)$ be distinct positive integers with $\gcd(V) = 1$ (n moving runners plus the stationary observer; the instances of this paper have $n \geq 3$, the range where the quotient geometry of §2 is nondegenerate). Runners start together at the origin of the unit circle $\mathbb{T} = \mathbb{R}/\mathbb{Z}$; runner p occupies $v_p t \bmod 1$ at time t. Write $\|x\|$ for the distance of a real number to the nearest integer. The *loneliness* of the stationary observer at time t is $g(t) = \min_{1 \leq p \leq n} \|v_p t\|$, and $M(V) = \max_{t \in \mathbb{T}} g(t)$. The Lonely Runner Conjecture, posed by Wills in 1967 and independently by Cusick in the language of view obstruction [1, 2, 3], asserts $M(V) \geq 1/(n+1)$ for every V. It is proved for up to fifteen runners: the classical cases $n \leq 7$ [1], eight runners [4], nine and ten [5], eleven through thirteen [6], and fourteen and fifteen [7]; beyond fifteen only computational verifications of special families are known [1, 8].

Attacking such a conjecture through equivalent rewrites is an old habit. The reformulation landscape around the Lonely Runner problem is unusually rich: Wills' original inhomogeneous-approximation form; Cusick's view obstruction; Rifford's continuous covering problem in dimension $n-2$, equivalent to a time-bound version of the problem [9]; and, from the program this paper reports on, a finite form—the pair-sum lattice reduction—under which the conjecture becomes a family of covering statements about explicit cyclic groups [1, 11]. Each rewrite is *lossless*: it holds exactly when LRC_n holds, with no residual error term. Each was adopted in the hope that its native toolkit—covering combinatorics, spectral theory, harmonic analysis, polyhedral geometry, Ehrhart theory—could move the conjecture.

This paper records, in a form we hope is useful beyond this one problem, what that hope runs into. The headline is a sharp negative observation, exhibited in four settings across two reformulation worlds:

> *LRC admits many lossless reformulations, and every natural strengthening of them that this program examined and exhibited—every reformulation that would open a new toolkit—is false. This is not a limitation of one technique; it is a structural pattern, documented in four settings (three manifestations of the pair-sum cascade and one zonotope–Ehrhart setting) with finite exact witnesses.*

The three cascade settings are not independent experiments: they are three probes of the same reformulation (an inventory audit, an instrument closure, and a rigidity falsification), and the paper counts them separately only because each fails for a different reason. Independent of the cascade is the fourth setting, the zonotope; the paper's claim is the pattern across both worlds, not four unrelated facts. The pattern claim is deliberately larger than the paper's evidence: four documented instances from one program are one thread's worth of confirmation, offered as a case study of a structural law that should hold verbatim in any comparable reformulation culture—lossless equivalence preserves hardness exactly, and the natural strengthenings fail where they matter. The thread's specific contributions are the pair-character identity of §5 (a small, proved, exact law that governs the arithmetic of the deepest holes) and the certified archive around it; the walls themselves are the general phenomenon.

The mechanism, which we call *type mismatch* and develop in §3, has three faces. First, losslessness transfers truth exactly: a proof of the reformulated statement is ipso facto a proof of LRC, and conversely. A rewrite may still change tractability—the program's own finitization is an equivalent rewrite that made the problem computable instance by instance—but it cannot discard the conjecture's content, and no lossless rewrite can be simultaneously strictly stronger than LRC and true at the extremal instances. Second, the statements that would make a reformulation tractable—a uniform margin, a spectral gap, a saturated bound with slack, a monotone scale law—are strictly stronger than LRC (they imply it; the converse fails); at the extremal instances the extra margin they demand does not exist, and they are false there with exact witnesses. Third, the instruments that a lossless reformulation naturally supports (classifications, invariances, exact counts) compose into statements of the wrong type: they close everything except the one gap that matters. All four settings were probed by exact computation, and the witnesses are small, checkable, and archived (§5, SI 8 and 9).

1.2 The four instances in one paragraph each

Instance 1: the finite covering reformulation (§3.2). The pair-sum equality theorem reduces $M(V)$ exactly to a maximum over cyclic groups $\mathbb{Z}_{v_u+v_w}$ (§4), turning LRC_n into a covering question about explicit bad sets. The program's classification toolkit (size laws, moat counts, class-group reductions, lift laws) closed 75.6% of a 3,712,576-instance corpus outright and localized the rest to one named class—but a statement-by-statement audit of the full proof inventory found *zero* proved statements of the missing type (a uniform growth-in-k margin over a trivial baseline);

every proved statement is a closure, invariance, exhaustiveness, or machine verification, and the one candidate that reads like an exception (the cut law's "k-cancellation") is a *k-invariance*—the exact opposite of a growth-in-k margin.

Instance 2: the transfer-matrix route (§3.3). The single instrument shaped to produce the missing type—a transfer-matrix or automaton model of the solution cascade, whose natural output is a spectral quantity uniform in scale—was executed as a proof attempt under a pre-committed success criterion and closed by theorem *in its single-anchor scope*: every single-anchor constrained transfer matrix of the lossless cascade is a partial permutation with spectral radius exactly 0 or 1; the cyclic product is diagonal, its trace is the survivor count itself, and at the six reflection-symmetric committed families the survival fraction is exactly 1 forever. The general branching strand-census operator is not covered by the theorem and is left open. Losslessness forbids generated decay for the single-anchor operator: bijections preserve measure, deletion filters without dissipating, and the single-anchor reformulation is bounded by its own fidelity.

Instance 3: character rigidity (§3.4). The harmonic-analytic route phrased the obstruction in terms of a closed system of ten characters. Its natural strengthenings—orbit-invariance of singleton or pairwise marginal profiles—are *false*, falsified by exhaustive exact computation; the marginal envelope saturates (is full) exactly at the configurations where the margin is needed, and the only invariant sparsity is joint. The trivial bound sits exactly at the cascade threshold: beating it by the factor the obligations require ($\sim$250 at every rung) is precisely the open margin, stated twice.

Instance 4: the Lonely Runner zonotope (§3.5). The quotient torus $\mathbb{R}^n/(\mathbb{R}v + \mathbb{Z}^n)$ carries a covering radius $\rho(v)$ in the quotient ℓ_∞ norm, and the Lonely Runner statement for the center class is $\delta(v) \le (n-1)/(2(n+1))$ with $\delta \le \rho$ always. The covering-radius form—*every* hole within the threshold, not just the center—would open the geometry-of-numbers toolkit. It is false from $n = 5$: at the extremal harmonic instance $v = (1, 2, 3, 4, 5)$ the covering radius is exactly $\rho = 16/47 > 1/3$ (a two-sided certificate), at $v = (1, 2, 3, 4, 5, 12)$ there is an argmax-certified point at $17/47 > 5/14$, at $v = (1, 2, 3, 4, 5, 6)$ one at $29/80 > 5/14$, at the non-extremal $v = (1, 2, 3, 4, 5, 7)$ one at $4/11 > 5/14$, and at $v = (1, 2, 3, 4, 5, 9)$ one at $1793/5000 > 5/14$. Meanwhile the Ehrhart h^*-polynomials of the zonotopes are beautifully behaved—real-rooted, log-concave, Newton, uniformly, through $n = 10$ in both core families—but their strictness decays like $\Theta(1/n)$ and has no measurable correlation with the covering margins: the coefficient-level toolkit cannot see the metric hole depth at all.

1.3 What this paper is and is not

The paper is deliberately plain about its shape. It is a *sharp negative result*—a whole class of approaches to a famous problem, and the exact reason each fails—plus *one bounded positive theorem* carried over from the companion monograph (the equality theorem, the rigid-zone classifications, and the p^2/pq composite structure, §4) and a *finite witness archive* with an explicit certification ladder (§5). Negative results of this kind are genuinely valuable when they come with exact computational evidence and a mechanism; the mechanism here (losslessness versus strengthening) applies verbatim to any problem with a comparable reformulation culture.

The boundary of the claims is as follows. The paper does not solve, reduce, or weaken the Lonely Runner Conjecture; no new cases of LRC are proved at any n; the rung results of [4, 5, 6, 7] are untouched. The equivalent forms that survive our audits (the δ-form of the covering statement, the pair-sum formulation) are exactly as hard as LRC, and we claim no leverage on them. The falsified strengthenings are stronger than LRC by construction, so their falsity implies nothing about LRC itself—which remains open, tight at the harmonic instances, and apparently indifferent to every toolkit this program could bring to bear. What the archive adds is certainty about where *not* to dig: four doors, each closed by exact computation or by theorem, each closing for the same reason.

1.4 Roadmap and conventions

§2 fixes notation and the four equivalent forms of the covering statement, with the proofs of equivalence deferred to the companion monograph where they are long. §3 states the type-mismatch mechanism and develops the four instances. §4 summarizes the positive core with exact statements. §5 presents the witness archive and the certification ladder, including the exact value $\rho = 16/47$ (certified from both sides), the argmax-certified $17/47$, the closed ladder row $m = 10$ ($\rho = 7/22 = \delta$), and the weight identity that governs their denominators. §6 concludes. SI 7 records the h^*-observations (the extension to $n = 9, 10$ is included here). SI 8 is the provenance ledger: every number cited in this paper is re-asserted by independent exact computation in a persisted, machine-checkable run (225 items, zero failures), and the certification scripts, budgets, and outputs are listed. SI 9 is the self-contained verification core: the elementary identity behind the four forms, the runner formulation with its Lipschitz certificate, the explicit pair-sum statement of Theorem 4.1, and one-line hand checks of the headline witnesses, so that a reader with only this paper can verify the sharpest negative claims without the companion monograph.

All computations are exact (rational arithmetic) unless explicitly marked as a float enclosure; the one exception class is flagged where it occurs (numerical root finding is supplemented by exact Sturm-sequence verification, and the statistical statements carry their sample sizes and p-values). Certification levels are labeled *exact* (complete exact proof), *certified* (complete proof with an independent numerical outer loop), or *consistent* (search-level evidence only), following the ladder of §2 [11].

2 Preliminaries

2.1 Notation

Throughout, $n \geq 3$ is the number of speeds, $V = (v_1, \ldots, v_n)$ has distinct positive integer entries with $\gcd(V) = 1$ and $v_1 = 1$ when a coordinate model is in use, and $\|x\| = \min(x - \lfloor x \rfloor, \lfloor x \rfloor + 1 - x)$ is the distance to the nearest integer. For a modulus N we write $\|x\|_N := \min_{j \in \mathbb{Z}} |x - jN|$ for the distance from x to the nearest multiple of N (so $\|x\|_N = N\,\|x/N\|$). We write $\lambda(V) = \max_{t \in \mathbb{T}} \min_p \|v_p t\|$ for the optimal loneliness and $M(V)$ for the same quantity; the two notations are interchangeable and we keep λ in metric contexts and M in the pair-sum context. The threshold is $\mathrm{thr}_n = (n-1)/(2(n+1)) = 1/2 - 1/(n+1)$. Two speed families recur: the *harmonic* family $V = (1, 2, \ldots, n)$ and the *rung-2* family $V = (1, \ldots, n-1, 2n)$. For these, λ is classical: $\lambda(1, \ldots, n) = 1/(n+1)$ exactly (the harmonic family is extremal, with zero margin against the conjecture), and $\lambda(1, \ldots, n-1, 2n) = 2/(2n+1)$ exactly (the rung-2 family is not extremal; its margin is $1/((n+1)(2n+1))$). Both identities are re-asserted exactly for $n \leq 10$ in the provenance run of SI 8.

2.2 Four forms of the covering statement

Write $\delta(V) = \frac{1}{2} - \lambda(V)$ and let $\mathbf{1} = (1, \ldots, 1)$. The following forms of LRC_n are used interchangeably in the literature and in this program, and it is essential to keep their strengths apart.

1. **Loneliness form.** $\lambda(V) \geq 1/(n+1)$ for all V.
2. **Center-depth form (δ-form).** $\delta(V) \leq \mathrm{thr}_n$, where $\delta(V) = \mathrm{dist}_\infty(\mathbb{R}v,\ \mathbb{Z}^n + \frac{1}{2}\mathbf{1})$ is the ℓ_∞ distance from the trajectory line to the shifted lattice. The identity $\delta = \frac{1}{2} - \lambda$ is elementary: for every t, $\max_j \|\frac{1}{2} - v_j t\| = \frac{1}{2} - \min_j \|v_j t\|$.
3. **Quotient-torus form.** Let $\mathcal{X}_v = \mathbb{R}^n/\mathbb{R}v$ be the quotient vector space (the calligraphic symbol avoids colliding with the speed vector V), let $\pi\colon \mathbb{R}^n \to \mathcal{X}_v$ be the projection, $\Lambda = \pi(\mathbb{Z}^n) \cong \mathbb{Z}^{n-1}$, and let $\|\cdot\|_{\mathcal{X}_v}$ be the quotient norm $\|c\|_{\mathcal{X}_v} = \mathrm{dist}_\infty(x, \mathbb{R}v)$ for any lift x of

c. Then $\delta(V) = \mathrm{dist}_{\mathcal{X}_v}(c^*, \Lambda)$ where $c^* = \pi(\frac{1}{2}\mathbf{1})$ is the *center class*, a 2-torsion point whose coordinates are $\frac{1}{2}$ on even speeds and 0 on odd speeds. The δ-form reads: the center class has quotient-norm depth at most thr_n in every $\mathcal{X}_v$.

4. **Covering-radius form (ρ-form).** Let $\rho(V) = \max_{c \in \mathcal{X}_v} \mathrm{dist}_{\mathcal{X}_v}(c, \Lambda)$ be the covering radius of Λ in the quotient norm. The ρ-form of the conjecture asserts $\rho(V) \le \mathrm{thr}_n$ for all V.

Forms (1)–(3) are equivalent to LRC_n: this is the content of the shift identity above together with the standard quotient-torus dictionary, and the equivalence is asserted computationally, exactly, on every instance of the program (the geometric pipeline is validated end to end in the runs of SI 8). Form (4) is *strictly stronger*. Since the maximum defining ρ ranges over all classes and δ is the depth of one class, $\delta(V) \le \rho(V)$ always; hence $\rho \le \mathrm{thr}_n$ implies the δ-form and thus LRC_n. The converse fails, and the failure is not an artifact of small examples: it begins exactly at $n = 5$ and persists (§3.5, §5). At the harmonic instances the δ-form is tight ($\delta = \mathrm{thr}_n$), so the ρ-form has no slack to absorb the difference between the center and the deepest hole: the two statements coincide on all tested instances for $n \le 4$—more precisely, the ρ-form *holds* on every instance tested at $n \le 4$, certified through $m = 24$ at $n = 3$ and $m = 16$ at $n = 4$, while the center-is-deepest property holds exactly at the multiples—which is why the identification was easy to believe—and separate from $n = 5$ on, with exact witnesses.

For computations we use two further standard facts, both proved and machine-validated in the program's records [11]. First, in *c-coordinates* $c_j = x_j - v_j x_1$ ($j = 2, \dots, n$; valid when $v_1 = 1$) the lattice is exactly $\mathbb{Z}^{n-1}$ and the quotient norm is the maximum of the finitely many linear forms

$$\phi^{1j} = \frac{c_j}{1 + v_j}, \qquad \phi^{ij} = \frac{v_j c_i - v_i c_j}{v_i + v_j} \quad (2 \le i < j),$$

each of ℓ_1-norm at most 1; hence $\|c\|_{\mathcal{X}_v} \le \|c\|_\infty$ and Lipschitz certification needs no norm-equivalence constant. Second, the depth function admits the one-parameter *runner formulation*

$$D(c) \;=\; \min_{\sigma \in [0,1)} \max\Big(\|\sigma\|,\; \max_{j \ge 2} \|c_j - v_j \sigma\| \Big), \tag{2.1}$$

which is the distance of c to Λ on the quotient torus; this is the form used by the branch-and-bound certifier of §5, and it was validated against the independent linear-form/lattice-search implementation on random rational points for every instance used in this paper (forty points per instance, exact agreement throughout).

2.3 The Lonely Runner zonotope and its Ehrhart theory

The *LR-zonotope* of V is $Z(V) = \pi([0,1]^n) \subset \mathcal{X}_v$; it is an $(n-1)$-dimensional lattice zonotope with respect to Λ, with relative volume $\sum_j v_j$ (normalized volume $d!\,\sum_j v_j$, so that $h^*(1) = d!\,\sum_j v_j$) [16, 12]. Its lattice-point geometry encodes the covering problem above: holes of the Λ-covering correspond to classes of large quotient depth. Its Ehrhart theory is arithmetic and clean [12, 13, 15]. Writing $d = n - 1$, the Ehrhart polynomial of $Z(V)$ is

$$\mathrm{Ehr}_{Z(V)}(t) \;=\; \sum_{k=0}^{d} e_k\, t^k, \qquad e_k \;=\; \sum_{\substack{S \subseteq [n] \\ |S| = k}} \gcd\big(v_{S^c}\big), \qquad e_d = \sum_j v_j, \tag{2.2}$$

and the h^*-polynomial is obtained from the Eulerian expansion $\sum_{t \ge 0} \mathrm{Ehr}(t) z^t = h^*(z)/(1-z)^{d+1}$; explicitly $h^*(z) = \sum_k e_k\,(1-z)^{d-k} A_k(z)$ with A_k the Eulerian polynomials in the shifted convention $A_0 = 1$, $A_1 = z$, $A_2 = z + z^2$ (so $A_k(z) = \sum_j \left\langle {k \atop j} \right\rangle z^{j+1}$ for $k \ge 1$) [17]. Formula (2.2) was derived in the program's records and verified in three independent ways on every instance used here (the Eulerian expansion, the standard Ehrhart inversion, and a depth-first coset count

at $t = 1, 2$; SI 8). The gcd structure is that of the arithmetic matroid of the projected basis [13], which is why the arithmetic-matroid literature is the natural home for the residual positive question in SI 7.

2.4 The certification ladder

Every classification claim in this paper carries one of three labels. *Exact*: a complete proof in rational arithmetic (the $d = 2$ arrangement enumerations; all refuting witnesses, which are single evaluations of D at rational points). *Certified*: a complete proof in which an outer numerical loop (a grid or a branch-and-bound frontier) is combined with an exact inner analysis and a rigorous one-sided error bound (the $d = 3$ away-certifications; the branch-and-bound certificates of §5). *Consistent*: search-level evidence only (grid maxima equal to the claimed value, multi-start optimization finding nothing above it), with no completeness claim. The labels exist precisely because enclosure-level evidence must not be presented as proof; the ladder is applied uniformly to every claim in this paper.

3 The type-mismatch mechanism and four instances

3.1 The mechanism

Call a statement R a *lossless reformulation* of LRC_n if R holds for all speed vectors at rung n if and only if LRC_n does, with no slack in either direction. Call a statement S a *toolkit strengthening* of R if (i) S implies R, strictly, and (ii) the strictness is of the kind that a named toolkit can exploit: a uniform margin over a trivial baseline, a spectral gap, an interior rate, a monotone law in the scale parameter, a saturated bound with slack. The distinction is not aesthetic; it is the difference between statements that classification, spectral, harmonic, or polyhedral instruments can consume and statements they cannot.

Three observations, jointly verified four times in this program, make up the mechanism.

1. **Losslessness transfers truth exactly.** A proof of the reformulated statement is ipso facto a proof of LRC_n, and conversely. A rewrite may still change tractability—the program's own finitization is an equivalent rewrite that made the problem computable instance by instance—but it cannot discard the conjecture's content: no lossless rewrite can be simultaneously strictly stronger than LRC_n and true at the extremal instances.
2. **The strengthening is false where it matters.** The toolkit-relevant strengthenings are strictly stronger than LRC (they imply it; the converse fails), and at the extremal instances (the harmonic and rung-2 families, the reflection-symmetric cascade families, the tight covering configurations) the extra margin they demand does not exist. When the demand is a definite statement, it is false, and small n exhibits the falsity with exact witnesses.
3. **The instruments land on the closed side.** The instruments a lossless reformulation naturally supports—exact counts, classifications, invariances, exhaustiveness checks—compose into statements that close everything *except* the missing margin: by theorem in Instance 2 (Lemma 3.1), by audit in Instance 1 (the inventory classification), and by measurement in Instances 3 and 4 (the saturation and decay records below).

The four subsections that follow develop one instance each. Three of them sit on the same pair-sum cascade (an inventory audit, an instrument closure, a rigidity falsification) and the fourth is independent of it (the zonotope); what is common to all four is the mechanism, and we state it once, here, as the paper's organizing claim: *in each setting, the falsity or unavailability of the strengthening is not an accident of the example but a consequence of the losslessness that made the reformulation faithful.*

3.2 Instance 1: the finite covering reformulation

The pair-sum equality theorem (§4) reduces $M(V)$ to a finite maximum over cyclic groups: $M(V) = \max_{\{u,w\}} \tau(u, w)$, where τ is the optimal loneliness on the pair-sum lattice of denominator $N = v_u + v_w$. Under this reduction LRC_n becomes the statement that for every V some pair's bad sets $B_p = \{k \in \mathbb{Z}_N : (n+1)\|v_p k\|_N < N\}$, $p \in [n] \setminus \{w\}$, fail to cover $\mathbb{Z}_N$: a finite covering problem, exact, and apparently made for the covering toolkit.

The toolkit delivered. The program proved an exact size law for the bad sets, a polygonal witness lemma, moat-counting and coincidence-counting lemmas, a lift law, and a class-group reduction at prime moduli; it classified the unit covering configurations completely at the rigid primes (Theorem 4.2); and on the full corpus of 3,712,576 primitive 5-sets with speeds ≤ 56 it closed 75.6% of instances by proved lemmas alone (78.2% with the coincidence lemma) and all of them by per-modulus verification, localizing the residual obstruction to a single named class [11]. What it did not deliver, at any point, is a single statement of the type the residual obstruction requires: a *uniform growth-in-k margin over a trivial or measured baseline.*

This is a verifiable claim, and it was verified by audit: the full proof inventory of the program—34 named results across the pair-sum, group-ring, tensor, and cascade sections—was classified statement-by-statement. Every proved statement is a closure, an invariance, an exhaustiveness, or a machine verification. The one candidate that reads like an exception, the cut law's "k-cancellation," resolves as a k-scale *invariance* of the composed bound: it cancels exactly the k-dependence that a margin would need to grow—the opposite of the missing type. The spacing/packing lemma, the chain-class volume law, and the dilate theorem were each examined as potential margin carriers and each resolves the same way: a k-growing *count* whose margin over its own trivial baseline is k-free, a degrading order-factor slack, or a conditional consequence of the hypothesis stack it was meant to prove.

The empirical half of the reading ("no proved margin-type statement exists in the inventory") is thereby settled. The structural half ("no classification toolkit *can* produce one") is a hypothesis about tool reach; it is what Instances 2–4 test, in three independent settings, and what the falsity archive of §5 completes on the metric side.

3.3 Instance 2: the transfer-matrix route, closed by theorem

The one margin-shaped candidate instrument in the entire program was a transfer-matrix (automaton) model of the solution cascade: transfer operators naturally emit spectral quantities uniform in the scale parameter, and a uniform spectral gap would be exactly the missing type. The route was built under a success criterion fixed before computing: the whole-cascade survival fraction must be bounded by ρ^{depth} times the initial mass, with spectral radius $\rho \leq 1 - \epsilon$ and ϵ independent of scale.

Lemma 3.1 (Monomial dichotomy; Lemma M of the program records, single-anchor scope)**.** *Every constrained transfer matrix of the lossless cascade* that tracks a single anchor placement along the drift *is a partial permutation matrix. Its spectrum is* $\{0\}$ *together with roots of unity; its spectral radius is exactly* 0 *or* 1*, never interior, at any scale. The affine pair maps compose to the identity around any closed fiber loop (the dilation ratios telescope), so the cyclic transfer product is diagonal, and its trace is the cascade's own survivor count.*

The proof is short once the state space is fixed: the cascade is non-consuming (pure intersection semantics), the drift maps are bijections, and a bijection constrained to an invariant subset is a partial permutation. *The scope is load-bearing*: the theorem covers the single-anchor operator only; the general strand-census operator scans the complement interval position by position with *branching* local coverage states, is not a partial permutation, and may have interior spectral radius (the 2×2 witness below, radius 1.122). The general case is left open; the single-anchor dichotomy is what is established. The machine record is not short: 1500/1500 random constrained systems;

12,000 exact trace fingerprints over 300 systems; 66 homogenized pair systems on the committed families, with spectral radius 1 at exactly the six reflection-symmetric families (their deletion sets close into 2-cycles and the two-fiber survival fraction is exactly 1.0000 at all depths) and 0 at the other sixty (total extinction within three iterates, the intermediates set by lattice runs of solution points, not by geometric decay). The criterion is unattainable by theorem for the single-anchor operator, not merely unattained.

Two further doors were checked. Coarse-grained or weighted automata can carry interior spectral radii numerically, but the radius dominates every cycle's geometric mean, so a uniform gap would force uniform decay of the census weights—the margin as an *input*, not an output—and the converse fails (a 2×2 witness with all cycle means ≤ 0.9 and radius 1.122: branching resurrects). Averaged or randomized operators collapse to the independent-extension baseline. And the reduction closes the loop: the whole-cascade survival fraction equals the product of consecutive conditional ratios, each bounded by the pair-volume and solution-set thinness hypotheses the route was commissioned to prove. The instrument is the census in different notation; it consumes the hypotheses it was built to generate. Losslessness forbids generated decay *for the single-anchor operator*—bijections preserve measure, deletion filters without dissipating—and the single-anchor reformulation is bounded by its own fidelity. The general branching strand-census operator is not covered and is left open.

3.4 Instance 3: character rigidity and saturated envelopes

The harmonic-analytic route phrased the cascade obstruction in terms of *characters*: the four singleton coordinates and six pairwise differences of the placement vector form a closed system of ten functionals (Lemma C10 of the program records), closed under the renormalization action up to affine shifts, whose aggregate marginal-multiset is an orbit invariant. Any class-level census law must live on this system; the route's natural strengthenings are laws about its per-character profiles.

They are false, exactly. The singleton-profile orbit invariance (Lemma V$'$) fails at both fully-enumerated primes: per-key counts match across orbit members only after a size permutation, and the profiles differ. The pairwise-profile invariance (V$''$) fails with 0/32 sigma-matched tuples. The mechanism is *anchor mixing*: the renormalization bijection couples every moving slot to the anchor coordinate; slot singletons map to anchor pairs and pairwise differences to non-anchor singletons, so no per-character statistic can be invariant—only the aggregate is.

Worse, the per-character envelopes are *saturated* exactly where the margin would be needed. On the committed top orbits the pinning envelope equals the modulus at both $k = 2$ primes and 27/29 at $k = 3$ (fullness fractions 1.000, 1.000, 0.931): the marginal boxes are full, the γ-envelope sits at or above the threshold value 4 at $k = 2$, and the only invariant sparsity is *joint* (median joint densities in the marginal box: 1.5%, 7.4%, 6.9%). A bound phrased on individual characters is tight—zero slack—precisely at the configurations the obligations target.

Finally, the trivial bound sits exactly at the cascade threshold. The baseline $|Sol| \leq p^{m-1}$ realizes the constant $c_1 = (T/(T-6) + o(1))^{m-1}$ (here and below $m = T - 3$ is the arc count of the frontier cell, so $m - 1 = T - 4$)—numerically 256, 243, 244.14 at $T = 8, 9, 10$—which is exactly the value at which the depth denominator of the conditional scaling theorem vanishes. The hypothesis H1g's content is to beat the trivial bound by a factor growing in k; the measured factors (95×, 1,207×, 3,803× at $k = 2, 3, 4$) grow, but the first of them starts *below* the threshold constant ($95 < 244$, the $k = 2$ factor against the $T = 10$ threshold $c_1 = 244.14$), which is precisely why the margin must grow in k and why $k = 2$ alone cannot certify anything; the proved factors do not exist, and the sharp-constant question and the margin question are one criterion stated twice. This is the type mismatch in its purest form: the character system is exactly rich enough to locate the obstruction and exactly too poor to bound it.

3.5 Instance 4: the Lonely Runner zonotope

The fourth instance is metric, and its falsity archive is the paper's sharpest. The quotient-torus model of §2 gives two statements: the δ-form (the center class is within thr_n of the lattice), which is equivalent to LRC_n, and the ρ-form (every class is), which is strictly stronger and would open the geometry-of-numbers toolkit—transference bounds, covering radius inequalities, dilution arguments. The natural lemma a covering-radius toolkit wants first is exactly $\rho \le \mathrm{thr}_n$.

It is false, with exact witnesses, from $n = 5$, and the two sharpest witnesses carry certified argmax structure. At the extremal harmonic instance $v = (1, 2, 3, 4, 5)$ the deepest hole has the exact depth

$$\rho \;\ge\; \frac{16}{47} \;>\; \frac{1}{3} = \mathrm{thr}_5,$$

attained at the rational point $c = (\frac{122}{1175}, \frac{688}{1175}, \frac{284}{1175}, \frac{80}{1175})$, whose deepest-hole structure binds four pair forms of the quotient norm under a weight identity that forces the denominator (§5.2): the deepest hole is not the center, and the covering-radius statement fails at the very instance where the δ-form is tight. At $n = 6$ the harmonic witness $c = (0, \frac{5}{16}, \frac{15}{16}, \frac{3}{4}, \frac{1}{2})$ gives $D = 29/80 > \frac{5}{14} = \mathrm{thr}_6$, and the non-extremal instance $v = (1, 2, 3, 4, 5, 7)$—whose center depth is the *computed* $\delta = 1/3$, strictly below thr_6 with slack—carries $c = (\frac{1}{16}, \frac{9}{16}, 0, \frac{3}{8}, 0)$ with $D = 4/11 > \mathrm{thr}_6$: the falsity is not an extremal-curiosity confined to the tight family, and the covering-radius failure there is not an artifact of center-class tightness. The rung-2 instance $v = (1, 2, 3, 4, 5, 12)$ has the exact witness $\rho \ge 17/47 > \mathrm{thr}_6$, attained at $c = (\frac{4714}{5875}, \frac{1}{8}, \frac{5}{8}, \frac{6}{25}, \frac{2034}{5875})$ (its earlier recorded witness sat at exactly $5/14 = \mathrm{thr}_6$, a zero-slack boundary case that separated the deepest hole from the center, $\delta = 9/26 < 5/14$, without strictly exceeding the threshold; the new witness upgrades it to a strict refutation), and the instance $v = (1, 2, 3, 4, 5, 9)$, which sits at the threshold from the center side ($\delta = 5/14$ computed), carries the strict witness $\rho \ge 1793/5000 > \mathrm{thr}_6$. All witnesses are non-torsion points—the deepest holes are not the center classes—and all are single exact evaluations of (2.1), checkable by hand (SI 9).

For $n \le 4$ the ρ-form holds on every instance tested: at $n = 3$ the deepest-hole property (that the maximum of D is attained at the center class) holds exactly when $3 \mid m$ in the family $(1, 2, m)$ for $m \le 24$, and at $n = 4$ it holds when $4 \mid m$ for $m \le 16$ (away-certified at $m = 4, 8, 12$; branch-and-bound certified at $m = 16$, §5.4), with exact witnesses against every other m. The two forms have the same truth value throughout the tested small range—which is precisely why the stronger form was tempting; the $n = 5$ witness is precisely why it is wrong.

The companion hope—that the zonotope's Ehrhart theory could control the covering radius through the positivity structure of its h^*-polynomial—fails independently, and its failure mode is the cleanest illustration of type mismatch in the whole program. The The h^*-polynomials are real-rooted, log-concave, and Newton, uniformly, in every instance tested through $n = 10$ in both core families (SI 7). But the log-concavity *strictness* S decays roughly like $30/n$ in the measured range (nS drifting from ≈ 37 to ≈ 30 in the harmonic family over $n = 3, \ldots, 10$, and from 60.5 to ≈ 30 in the rung-2 family), and Newton's inequalities force S above a floor $\approx 1 + 4/d$ that itself declines, so much of the decay is generic and the $1/n$ behavior cannot persist indefinitely—there is no growing quantitative margin to spend; and across the instance sweeps the strictness has no statistically distinguishable correlation with the covering margins at the achievable sample sizes (Spearman $+0.32$, $p = 0.32$, $N = 12$ at $n = 3$; -0.31, $p = 0.45$, $N = 8$ at $n = 4$; the sign flip is consistent with noise; these arrays are too small to establish correlation or its absence). A coefficient inequality of a counting polynomial at growing scale t is the wrong type to pin a fixed-scale metric hole depth; the computation says so twice, once by decay and once by the absence of a measurable signal, and the covering-radius statement it was meant to feed is false at the extremal instances anyway. There is no mechanism, and none is needed: the bridge is dead on both ends.

3.6 The pattern

Table 1 assembles the four instances. Each row names the reformulation, the toolkit strengthening it wanted, the status of that strengthening, and the witness. The rows are independent; the last column is always finite, exact, and small. We emphasize what the table does *not* contain: any statement falsifying LRC itself, or any statement weakening our certainty that the surviving equivalent forms (the δ-form, the pair-sum form) are exactly as hard as the original.

Table 1: The four instances of type mismatch. "Margin type" is the statement class the toolkit needed; "status" is its verified fate, with the kind of verification distinguished: *false* (an exact witness falsifies a definite statement), *closed by theorem* (a theorem shows the instrument cannot produce the output), and *zero in inventory* (an audit finding: no statement of the type exists among the proved results). The first three rows sit on the same pair-sum cascade; the fourth is the independent zonotope setting.

Reformulation	Toolkit wanted	Status	Witness
Pair-sum covering	growth-in-k margin	zero in inventory (audit; 34 statements); k-cancellation is a k-invariance	audit record (34 statements)
Transfer cascade	spectral gap $1-\epsilon$	closed by theorem (Lemma M; survival $= 1$ at reflection families)	six families, exact
Character rigidity	per-character profile laws	false (V'/V'' witnesses); envelopes saturated; trivial bound at threshold	exhaustive orbits; 0/32
LR-zonotope	$\rho \le \mathrm{thr}_n$ (covering)	false from $n = 5$; h^* strictness decays, no measurable signal	16/47, 17/47 (argmax), 29/80, 4/11, 1793/5000 (exact)

4 The positive core: classification at the rigid primes and at p^2/pq

The program's attacks were launched from a positive base, and that base survives the negative results intact. This section states it with precision; the full proofs, the verification records, and the measurement methodology are the subject of the companion monograph [11], and nothing

here improves or weakens them. What matters for this paper's thesis is that the positive core is exactly as bounded as it is: everything it proves is a closure, classification, or identity, and its one conditional theorem wears its hypotheses on its sleeve.

4.1 The sum-only equality theorem

The analytic folklore (recorded in a comment on Tao's weblog [10]) is that the optimal time is rational with denominator the sum or difference of two speeds. The sharpening the program proved is that *sums alone suffice.*

Theorem 4.1 (Sum-only equality; Theorem 1.1 of the companion monograph)**.** *Let* $V = (v_1, \ldots, v_n)$ *have distinct positive entries with* $\gcd(V) = 1$ *and* $0 < M(V) < \frac{1}{2}$*. For each unordered pair* $\{u, w\}$ *let* $\tau(u, w)$ *denote the optimal loneliness of the* $(n-1)$*-runner instance obtained by restricting to the pair-sum lattice of denominator* $N = v_u + v_w$*—explicitly (SI 9),*

$$\tau(u, w) \;=\; \max_{k \in \mathbb{Z}_N} \; \min_{p \in [n] \setminus \{w\}} \left\| \tfrac{v_p k}{N} \right\|$$

(runners u *and* w *coincide on the lattice, since* $\|v_u k/N\| = \|v_w k/N\|$ *there). Then*

$$M(V) \;=\; \max_{\{u,w\}} \tau(u, w),$$

and consequently LRC_n *holds for all* V *if and only if the finite statement* (TAU_n) *does: for every* V*, some pair-sum lattice carries an optimal value* $\geq 1/(n+1)$*. The finitization is exact; no information is lost, and nothing about the classical difficulty is claimed to become easier.*

The two ingredients are that the binding runners at a global optimum can be chosen two-sided, forcing the optimal time onto a *sum* lattice (difference denominators never carry the optimum), and a mirror lemma: on that lattice the two binding runners are exact reflections of each other at every lattice time, so the n-runner minimum there equals an $(n-1)$-runner minimum. The equivalence of Corollary form ($\mathrm{LRC}_n \iff \mathrm{TAU}_n$) is the lossless gateway through which every instance of §3.2 entered.

4.2 The covering framework and the rigid zone

Under the equality theorem, a pair *certifies* an instance unless its $n-1$ bad sets $B_w = \{k \in \mathbb{Z}_N : (n+1)\|wk\|_N < N\}$ cover $\mathbb{Z}_N$. The framework that follows is finite and exact, and its proved pieces include a size law for the bad sets, a polygonal witness lemma, moat counting, a coincidence-counting lemma, a lift law, and a class-group reduction at prime moduli (Lemma 2.2–2.7 of the companion monograph). Five empirically distinct obstruction classes from the program's early experiments collapse, in this language, into additive orders of one covering phenomenon.

Theorem 4.2 (Rigid-zone classifications; Theorems 3.1–3.5 of the companion monograph)**.** *At rung* $n = 4$ *the unit-capable moduli—those* N *for which* $n-1$ *unit bad sets can cover* $\mathbb{Z}_N$*—are* $7, 11, 13$*; at rung* $n = 5$ *they are* $7, 13, 17, 19, 37$ *within the verified range (primes* ≤ 150*, composites* ≤ 111*). At every one of these primes the covering configurations are classified completely by a short explicit list of translate classes in the cyclic group: antipodal classes and domino tilings at* 7 *and* 11*; perfect matchings at* 13*; the classes* $\{0, 1, 2, 7\}$*,* $\{0, 2, 4, 6\}$ *at* 17 *and* 19 *(together with one further class at* 19*); and the single class* $\{0, 2, 12, 15\}$ *at* 37*.*

This is the program's central structural finding: *at rigid primes, the covering obstruction is governed by a small translate-class structure*, and the classification is what closes 75.6% of the 3,712,576-instance corpus outright. The residual obstruction is localized (mixed order signatures at composite moduli), and the composite side carries the structure that the next subsection records.

4.3 The composite side: group ring, tensor, and the fiber cascade

Beyond the unit capable moduli, at $N = p^2$ and $N = pq$, the program built the composite-side structure that the scaling question lives on. The group-ring translation expresses the covering count as a cyclotomic coefficient problem; the pq side decomposes as a tensor problem whose Lam–Leung-type structure [18] fixes the admissible fiber-count signatures; and the foot-count input to the cascade was discharged by an exact identity verified on 10,410 triples with the fixed threshold at $p = 31$. The higher rungs carry a general-T fiber machinery, an exact dilate-intersection lemma, a reflection-fiber invariance, a chain-class dilate theorem, and a renormalization lemma; the Mirsky–Newman covering-system theorem and the Rédei–de Bruijn–Schoenberg structure theorem for vanishing sums of roots of unity [19, 18] play the classical supporting roles.

Everything above is proved and machine-verified. What is *not* proved is the scaling closure itself, and the program consolidated it into a single conditional statement with a disclosed ledger.

Theorem 4.3 (Conditional scaling closure; Theorem 5.1 of the companion monograph, $\pm$-reduced)**.** *Assume the multi-fiber survivor inequality $Hq_{\pm}$ (with additive error term), the distinct-family volume law H1g, and the resonance zone cutoff H2r of the companion's ledger. Then the fiber cascade over the $\pm$-representative fiber set $U_{\pm}$ empties the frontier cell $(1K, T-3U)$ of every rung $T \geq 8$ at $N = p^2$, within a depth that is $O_T(\log k)$ (explicit leading term $\log k / \log(T/(T-6))$); no k-uniform linear bound is claimed, and the earlier "$\leq T + 3$" is withdrawn. The reduction chain and every other input are proved or machine-validated; each constant's provenance—derived, measured, or fitted, with the data it is fit to—is disclosed in the six-entry hypothesis ledger, and the open obligations are reduced to named objects: the strand census (for H1g), the sharp pair-line constants (for H2m), and the zone cutoff Λ_0 (for H2r). Cell emptiness is a covering statement only; it does not by itself imply* LRC *at the rung.*

We do not restate the ledger here. Its relevance to this paper is its shape: the proved side of the ledger is all closures and identities; the open side is exactly the growth-in-k margins whose type-mismatch career occupies §3. Theorem 4.3 is what the positive core amounts to when stated at its exact strength, and it is why the program's claim to a positive result is deliberately the bounded one: *the obstruction is completely classified where classification suffices, and the residual is a named margin of a type no turn of the program ever proved.*

5 The witness archive

The archive has four shelves: exact refutations of the covering-radius form (each a single rational point, checkable by one evaluation of (2.1), the sharpest carrying a complete two-sided certificate—the exact covering radius itself), the arithmetic law that governs the denominators of the exact values (§5.2), the deepest-hole certification ladder for the family $(1, \ldots, n-1, m)$, and the branch-and-bound certificate method together with its negative-control validation. Every entry is re-asserted by two independent exact implementations in the provenance run of SI 8.

5.1 Exact refutations of the covering-radius form

Table 2 lists the witnesses. Each witness c is a rational point of the quotient torus in c-coordinates; the claimed value is $D(c)$ from (2.1), computed exactly and independently by the linear-form/lattice-search implementation of the program's records. The first entry carries the strongest certification in the archive: the value $16/47$ is the covering radius itself, certified from both sides—the witness below, and the branch-and-bound upper certificate of §5.4, whose tree empties at $r = 16/47$. The second entry is *argmax-certified*: the displayed value is not merely attained at the witness point but is the maximum of D on the whole argmax neighborhood, certified exactly by the arrangement analysis of §5.4 (all vertices, ridges, and corners of the region evaluated in rational arithmetic), and the argmax carries the binding structure of §5.2 that forces

the denominator; its global equality remains open (§5.4). The harmonic instance is extremal for the δ-form ($\delta = \mathrm{thr}_n$ exactly), so the value 16/47 kills the ρ-form at the very instance where the surviving equivalent form is tight; the rung-2 instance $v = (1, 2, 3, 4, 5, 12)$ was previously a boundary row (witness value exactly thr_6) and is now a strict refutation; the $m = 9$ row is upgraded from boundary to strict witness; the $v = (1, 2, 3, 4, 5, 6)$ and $v = (1, 2, 3, 4, 5, 7)$ rows are strict witnesses of long standing. The witnesses are non-torsion in every case: the deepest holes are not the center classes.

Table 2: Exact witnesses against the covering-radius form $\rho(V) \le \mathrm{thr}_n$. All values are exact rational lower bounds for ρ; the first is *exact-certified* (the branch-and-bound tree empties at r equal to the displayed value, so the value is the covering radius itself), the second *argmax-certified* (the maximum of D on the whole argmax neighborhood equals the displayed value, exact arrangement analysis); both are depths of the deepest hole, not merely point depths. Witness points are in c-coordinates (§2). "> thr?" marks whether the value *strictly* exceeds the threshold: only strict rows falsify the ρ-form; boundary rows (equality) separate the deepest hole from the center class without falsifying it. The δ entries for non-harmonic instances are computed values, not harmonic identities.

V	$\delta = \frac{1}{2} - \lambda$	thr_n	witness c	ρ	> thr?
$(1, 2, 3, 4, 5)$	1/3	1/3	$(\frac{122}{1175}, \frac{688}{1175}, \frac{284}{1175}, \frac{80}{1175})$	$16/47 \approx 0.3404$	yes (exact)
$(1, 2, 3, 4, 5, 12)$	9/26	5/14	$(\frac{4714}{5875}, \frac{1}{8}, \frac{5}{8}, \frac{6}{25}, \frac{2034}{5875})$	$17/47 \approx 0.3617$	yes (argmax)
$(1, 2, 3, 4, 5, 9)$	5/14	5/14	$(\frac{1}{32}, \frac{15}{32}, \frac{1383}{10000}, \frac{32921}{40000}, \frac{95473}{200000})$	$1793/5000 = .3586$	yes
$(1, 2, 3, 4, 5, 6)$	5/14	5/14	$(0, \frac{5}{16}, \frac{15}{16}, \frac{3}{4}, \frac{1}{2})$	$29/80 = .3625$	yes
$(1, 2, 3, 4, 5, 7)$	1/3	5/14	$(\frac{1}{16}, \frac{9}{16}, 0, \frac{3}{8}, 0)$	$4/11 \approx .3636$	yes

The $m = 9$ witness deserves its sentence: the instance sits at the threshold from the center side ($\delta = 5/14 = \mathrm{thr}_6$ computed), so the boundary witness at exactly 5/14 could not decide the ρ-form either way; the strict witness $1793/5000 > 5/14$ does. What it does not decide is the value itself: the exact covering radius at $v = (1, 2, 3, 4, 5, 9)$ is unknown. The recorded lower bound is marginally deeper than the clean witness—an exactly evaluated dyadic search point at depth $0.358639\ldots$ persists in the provenance data (SI 8)—while no nontrivial upper certificate is on record. By convention the headline is the clean rational witness 1793/5000 (exact, hand-checkable); the dyadic point is recorded as a deeper provenance artifact and is not used as the headline. The row's status is strict refutation, exact value unknown.

Three further exact families fill out the record—but it matters what each refutes. The $n = 5$ family $(1, 2, 3, 4, m)$ refutes the *modular deepest-hole law*, not the ρ-form: the law's equality clause (deepest hole at the center when $5 \mid m$) fails exactly at the multiples $m = 5$ (now with the exact witness $\rho \ge 16/47 \ne \delta = 1/3$) and $m = 15$ ($6/19 > 5/16 = \delta$), while the deepest-hole-at-center property fails at all twenty non-multiples (exact witnesses, the largest margins $1/3 > 3/10$ at $m = 6, 7, 8$). The $n = 6$ family extinguishes the modular law at every tested $m \in \{6, \ldots, 11, 18, 24\}$ (all multiples of 6 through $4n$ included; the narrowest modular margin is 7/5200 at $m = 24$, witness value 71/208 against $\delta = 17/50$); against the ρ-form itself those witnesses fall short of the threshold ($6/19 < 1/3$; $71/208 < 5/14$) and are recorded as deepest-hole evidence, not covering-radius refutations. At $n = 6$, $m = 9$ the strict witness of Table 2 upgrades the earlier boundary row; at $m = 12$ the witness 17/47 upgrades another. A useful way to read the table: the ρ-form is falsified at the extremal harmonic instance by the exact argmax value 16/47, at a second $n = 6$ instance by the exact argmax value 17/47, and at two further $n = 6$ instances by exact strict witnesses; the classification side (which instances have their deepest hole at the center) is a different question, answered by the ladder below.

5.2 The arithmetic of the deepest holes: a weight identity

The exact values 16/47 and 17/47 are not accidents of the search: their denominators are forced by a one-line linear identity among the norm's defining forms, and the same identity explains why the exact covering radii of these instances are rationals with small structured denominators rather than generic reals. This subsection records the identity, the two binding archetypes that activate it, and the resulting denominator laws; everything is elementary and fully proved.

Recall from §2 that the quotient norm is the maximum of the pair forms ϕ^{uv}, and write the *pair character* $\chi_{uv} = (u{+}v)\,\phi^{uv}$, i.e. $\chi_{uv}(x) = v\,x_u - u\,x_v$ on any lift x: a circle-valued linear map on the quotient torus $\mathcal{X}_v/\Lambda$ (integer shifts of x change χ_{uv} by integers). For a lattice translate z, write $k_{uv}(z) = v\,z_u - u\,z_v \in \mathbb{Z}$, so that $\phi^{uv}(y-z) = (\chi_{uv}(y) - k_{uv}(z))/(u{+}v)$.

Lemma 5.1 (pair-character identity)**.** *For any three distinct speeds $a, b, c \in V$,*

$$c\,\chi_{ab} \;+\; a\,\chi_{bc} \;=\; b\,\chi_{ac} \tag{5.1}$$

identically on the quotient torus.

Proof. Substitute $\chi_{uv} = v\,x_u - u\,x_v$ (any lift): the left side is $c(bx_a - ax_b) + a(cx_b - bx_c) = bc\,x_a - ac\,x_b + ac\,x_b - ab\,x_c = b(cx_a - ax_c)$, which is the right side. Every term is a function of the quotient class, so the identity descends. □

Multiplying (5.1) through by the pair sums recovers the *weighted form* in terms of the pair forms themselves:

$$c\,(a{+}b)\,\phi^{ab} \;+\; a\,(b{+}c)\,\phi^{bc} \;=\; b\,(a{+}c)\,\phi^{ac}. \tag{5.2}$$

The weight of each pair is the third speed times the pair sum. Three pair forms of one speed triple are therefore never independent, and at a point where all three are simultaneously *binding*—each equal to $\pm\rho$ in absolute value relative to its own nearest translate—the identity becomes an arithmetic constraint on ρ itself.

Corollary 5.2 (denominator laws)**.** *Let y be a point of the quotient torus and $\rho > 0$.* (i) Triple binding. *If the three pair forms of a speed triple $a < b < c$ are simultaneously binding at y, $\phi^{uv}(y - z_{uv}) = s_{uv}\,\rho$ with signs $s_{uv} \in \{\pm 1\}$ and translates z_{uv}, then*

$$\big[\,c\,s_{ab}(a{+}b) \;+\; a\,s_{bc}(b{+}c) \;-\; b\,s_{ac}(a{+}c)\,\big]\,\rho \;\in\; \mathbb{Z},$$

the integer being $b\,k_{ac} - c\,k_{ab} - a\,k_{bc}$. In the sign pattern $(+,+,-)$ and its negation—the pattern that occurs at both closed argmaxes below—the bracket equals $2(ab{+}ac{+}bc)$, the sum of the three weights in (5.2), and

$$\rho \;=\; \frac{\big|\,b\,k_{ac} - c\,k_{ab} - a\,k_{bc}\,\big|}{2\,(ab + ac + bc)}.$$

In particular the reduced denominator of ρ divides $2(ab + ac + bc)$. (ii) Pair bisection. *If a single pair form (u, v) binds at two translates with opposite signs, $\phi^{uv}(y{-}z_1) = +\rho$ and $\phi^{uv}(y{-}z_2) = -\rho$, then $2(u{+}v)\,\rho = k_{uv}(z_2) - k_{uv}(z_1) \in \mathbb{Z}$: the reduced denominator of ρ divides $2(u + v)$.*

Proof. Part (i): binding means $\chi_{uv}(y) = k_{uv}(z_{uv}) + s_{uv}(u{+}v)\rho$ for each of the three pairs; substitute the three expressions into (5.1) and collect the ρ-terms. In the stated sign pattern the coefficient of ρ is $c(a + b) + a(b + c) + b(a + c) = 2(ab + ac + bc)$. Part (ii) is the same substitution for the single identity $\chi_{uv}(y){-}\chi_{uv}(y) = 0$ with the two translates: $k_{uv}(z_1){+}(u{+}v)\rho = k_{uv}(z_2) - (u{+}v)\rho$. □

The scope of Corollary 5.2(i) is worth stating precisely, because the harmonic example exceeds it. The hypothesis is that the three pair forms of *one* speed triple bind—nothing is assumed about the other pair forms, which may bind or not as they please. An argmax with four or

more binding forms therefore falls under the corollary once for each speed triple whose three pair forms all bind: the identity is applied to a three-element subset of the binding set, and additional binding forms neither disturb nor dilute it—every such triple with the $(+,+,-)$ sign pattern forces $2(ab{+}ac{+}bc)\,\rho \in \mathbb{Z}$ independently. The harmonic argmax below binds four pair forms; the triple $\{3,4,5\}$ among them forces $2 \cdot 47\,\rho \in \mathbb{Z}$, and the fourth binding form (the pair $(2,5)$) is extra structure the computation does not need. No comparable constraint exists at a point where fewer than three pair forms bind in the pattern; the $m = 9$ witness below is a chain of dips that never fully ties, and its clean value reflects the parametrized scan that found it rather than any forced arithmetic.

Both argmaxes with certified structure are triple-binding, and their architectures read as follows. At the harmonic instance $v = (1,2,3,4,5)$, the deepest hole $c^* = (\frac{122}{1175}, \frac{688}{1175}, \frac{284}{1175}, \frac{80}{1175})$ has four binding pair forms: $\phi^{45} = -\rho$ at $z = (0,0,0,-1)$, $\phi^{34} = -\rho$ at $z = (0,1,0,0)$, $\phi^{35} = +\rho$ at $z = (0,0,0,0)$, and $\phi^{25} = +\rho$ at $z = (0,1,1,1)$. The first three are the triple $\{3,4,5\}$ with signs $(-,-,+)$; Corollary 5.2(i) gives $2(3\cdot4{+}3\cdot5{+}4\cdot5)\,\rho = 5k_{34}{+}3k_{45}{-}4k_{35} = 5\cdot4{+}3\cdot4{-}4\cdot0 = 32$, i.e.

$$\rho \;=\; \frac{32}{2\cdot 47} \;=\; \frac{16}{47}, \qquad 47 = 3\cdot4 + 3\cdot5 + 4\cdot5.$$

The harmonic argmax set is richer than a single point, and the identity is the reason. Along the branch region containing the argmax, the three dips of the triple $\{3,4,5\}$ obey the weighted identity with *constant* total, $32\,r_{(3,5)} + 27\,r_{(4,5)} + 35\,r_{(3,4)} = 32$ (the c-terms cancel in (5.2)), so at every point of the region the smallest dip is at most $32/94 = 16/47$; on the one-parameter *balance family*

$$c(k) \;=\; \tfrac{1}{235}\,(2k{+}160,\; 3k{-}13,\; 4k{+}91,\; 5k), \qquad k = 20,\ldots,35,$$

the three dips tie at exactly $16/47$, so the bound is attained along a whole band. The family member $c(35) = (\frac{46}{47}, \frac{92}{235}, \frac{231}{235}, \frac{35}{47})$ is re-asserted exactly by the cross-validation run of SI 8 alongside the isolated argmax of Table 2; both are components of the argmax set, and the two-sided certificate of §5.4 covers them together. At $v = (1,2,3,4,5,12)$ the deepest hole $c^* = (\frac{4714}{5875}, \frac{1}{8}, \frac{5}{8}, \frac{6}{25}, \frac{2034}{5875})$ has three binding pair forms: $\phi^{2,12} = -\rho$ at $z = (1,0,0,0,-1)$, $\phi^{5,12} = +\rho$ at $z = (1,0,1,0,1)$, and $\phi^{25} = +\rho$ at $z = (1,1,2,2,4)$—the triple $\{2,5,12\}$ with signs $(+,+,-)$ in the $(ab),(bc),(ac)$ roles; the identity gives $2(10{+}24{+}60)\,\rho = 5k_{2,12}{-}12k_{25}{-}2k_{5,12} = 70{-}12{+}10 = 68$, i.e.

$$\rho \;=\; \frac{68}{2\cdot 94} \;=\; \frac{17}{47}, \qquad 94 = 2\cdot5 + 2\cdot12 + 5\cdot12.$$

(For the harmonic instance the same substitution reads $5k_{34} + 3k_{45} - 4k_{35} = 32$.) The weighted identity (5.2) is exactly the relation $35\,\phi^{34}{+}27\,\phi^{45} = 32\,\phi^{35}$ (resp. $84\,\phi^{25}{+}34\,\phi^{5,12} = 70\,\phi^{2,12}$) that the binding values satisfy; summed over the three binding pairs it reads $35r_{34}{+}27r_{45}{+}32r_{35} = 32$ (resp. $84r_{25} + 34r_{5,12} + 70r_{2,12} = 68$) with r the common binding value, which is how the two argmax values were first read off the computed deepest holes.

Three remarks sharpen the record. First, the lemma converts the deepest hole problem at such points into arithmetic: once the binding triple and its translates are known, ρ is a one-line integer computation, and the congruence class of the argmax ($\chi_{uv} \equiv s_{uv}(u{+}v)\rho \bmod 1$ for the binding pairs) turns the exact search into a finite parametrized scan—the method that located the $m = 12$ argmax above. Second, the repeated denominator 47 is not a law: $ab + ac + bc = 47$ for the triple $\{3,4,5\}$ and $ab + ac + bc = 94 = 2 \cdot 47$ for $\{2,5,12\}$, and the reductions by $\gcd 2$ and $\gcd 4$ happen to land on the same 47; at $v = (1,2,3,4,5,7)$ the recorded witness $4/11$ is pair-bisecting (Corollary 5.2(ii) with the pair $(4,7)$: $2 \cdot 11\,\rho = 8$), a different archetype with denominator governed by the pair sum 11, and no 47 appears—the denominator follows the binding architecture. What remains true, and worth recording, is that both computed triple-binding argmaxes share the 47: a coincidence the identity does not force (the two weight sums are 47 and 94, and the divisors 2 and 4 are incidental) but which the two data points do not demystify either. Whether the small-triple weight sums $ab + ac + bc$ cluster at the

scale where the deepest holes of small instances live is a question the archive legitimately raises and does not answer. Third, the archetypes have different reach: triple binding with weights $c(a+b)$, $a(b+c)$, $b(a+c)$ summing to $2(ab+ac+bc)$, and pair bisection with weight $2(u+v)$—and between them they govern every exact value in the archive. The four certified ladder values at the $n = 5$ multiples are pair-bisecting at the center class on the pair $(1, m)$—verified by the same binding-structure extraction, the pair binding at two translates with opposite signs at each of $m = 10, 20, 25, 30$—with reduced denominators 22, 42, 13, 62 all dividing $2(1+m)$; the two triple-binding argmax values divide $2(ab+ac+bc)$. The boundary rows at exactly the threshold (for example $m = 6, 7, 8$ at $n = 5$) bind neither archetype cleanly and are certified at their boundary values directly.

5.3 The deepest-hole certification ladder

Table 3 records the corrected modular law with its certification levels. The law as originally stated ("deepest hole at the center iff $n \mid m$") is a small-n phenomenon, genuine and exact at $n = 3$, certified at $n = 4$, broken at $n = 5$ by the harmonic instance itself—now with the exact value $\rho = 16/47$ against $\delta = 1/3$—and extinct at $n = 6$; the record below is the corrected version, and the rows improved by the branch-and-bound method are marked.

Table 3: The deepest-hole ladder for $(1, \ldots, n-1, m)$: when is the maximum of D attained at the center class ($\rho = \delta$)? "Exact" = complete exact proof; "certified" = complete proof with an independent outer loop; "consistent" = search-level only.

n	tested m	status of "deepest hole at center iff $n \mid m$"
3	3..24	exact: holds iff $3 \mid m$; exact witnesses against all others
4	4..16	certified: holds iff $4 \mid m$ ($m = 4, 8, 12$ away-certified; $m = 16$ certified by branch and bound); exact witnesses against all non-multiples
5	5..30	law false: $m = 5$ (harmonic; closed $\rho = 16/47 \neq 1/3$) and $m = 15$ fail exactly; $m = 10, 20, 25, 30$ certified by branch and bound ($\rho = 7/22$, 13/42, 4/13, 19/62 exactly, at the center class); all 20 non-multiples fail exactly
6	6..12, 18, 24	extinct: every tested m fails exactly, including every multiple of 6 through $4n$; at $m = 12$ the failure is the exact witness $\rho \geq 17/47 > \delta = 9/26$

The $n = 5$ multiples carry the hold signature (the torsion center attains δ; the $K = 32$ grid maximum equals δ exactly; a flat ridge passes through the center; a 288-start Nelder–Mead search with exact re-evaluation finds nothing above δ), but at $d = 4$ a global exact certificate was beyond the program's earlier arrangement machinery. (Only $m = 10$ is a rung-2 instance; $m = 20, 25, 30$ are the higher multiples of 5.) The branch-and-bound method of §5.4 closed all four. At $m = 25$ and $m = 30$ the trees empty in under five minutes each (10,414 and 10,646 certified leaves; $\rho = 4/13$ and $\rho = 19/62$ exactly), and at $m = 20$ the tree empties in sixteen minutes (37,018 leaves; $\rho = 13/42$). At $m = 10$ the first attempt (25 minutes, 143,228 box evaluations, 61,038 exact leaf certificates) did not terminate: the unresolved set had total volume below 6×10^{-7}, the exact depths at its sampled centers were at most $1135/3584 < 7/22$, and the largest exact U-bounds among the sampled survivors were all exactly $571/1792 = 7/22 + 9/19712$—a fixed quantization offset, consistent with pure minimax loss rather than genuine excess, though a residual volume below 6×10^{-7} does not exclude a deeper point hiding inside it. The record left two points open to objection—the search evidence behind the label was thin, and the residual reading was unproved—and both were answered in a renewed pass that closed the row. On the search side: 4,194,304 uniform random points, the full $\{k/16\}^4$ and $\{k/32\}^4$ dyadic grids, a 500,000-point torsion neighborhood, 2,048 batched random uphill climbs (1.8 million evaluations), and 64 Nelder–Mead polishes, all screened on a fixed s-grid over $[0, r] \cup [1 - r, 1)$—a *sound* screen, because the grid minimum of $G(c, \cdot)$ is an upper bound for $D(c)$, so no point above the threshold can escape it—followed by exact rational re-evaluation of the 114 deepest candidates:

every line of the search converges to the torsion center $(\frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2})$, at which the depth is exactly $7/22$, and nothing anywhere exceeds $7/22$. On the certificate side, the renewed cascade adds the exact Lipschitz box test of §5.4: at the first attempt's residual width ($1/512$ per side, half-width $1/1024$) the test closes the observed gap exactly, since $1135/3584 + 1/1024 = 0.31766\ldots < 7/22$, and the re-run tree empties (190,295 box evaluations, 95,147 splits, *zero* survivors, 1,286 seconds on one core; 846 of 846 sampled float-stage prunes re-confirmed in exact rational arithmetic; the harmonic negative control still refuses; the $n = 4$ certificates re-certify with identical box counts). Together with the torsion center attaining $7/22$, this certifies

$$\rho(1, 2, 3, 4, 10) \;=\; \tfrac{7}{22} \quad \text{exactly},$$

and since $7/22 = \delta = 1/2 - \lambda$ with $\lambda = 2/11$, the deepest hole at $m = 10$ is the center class itself. The $n = 5$ multiples residue now has a clean shape, recorded without forcing it into a law: the odd multiples $k \in \{1, 3\}$ fail exactly, the even and higher multiples $k \in \{2, 4, 5, 6\}$ are certified.

5.4 The branch-and-bound certificate and its negative control

The certification method is a second, independent construction, built on the runner formulation (2.1) rather than on the arrangement enumeration of the program's records. For a dyadic c-box C define

$$U(C) \;=\; \min_{\sigma \in [0,1)} \; \max\Big(\|\sigma\|,\; \max_{2 \le j \le n} \; \sup_{c \in C} \|c_j - v_j \sigma\| \Big),$$

the box-supremum analogue of (2.1) itself (the $\|\sigma\|$ term is the depth of the first runner, whose coordinate is fixed at 0; the inner maximum runs over $j = 2, \ldots, n$). It is computed exactly: each inner supremum is piecewise linear in σ with a rational kink set, so the minimum of their maximum is attained at a rational kink or crossing. By the minimax inequality, $\max_C D \le U(C)$, so a box with exact $U(C) \le r$ is *certified* to contain no point of depth above r; boxes that fail are split dyadically. If the tree empties, $\rho \le r$ is exactly certified by a finite, auditable leaf list. Two soundness refinements matter in practice: the σ-search is restricted to $[0, r] \cup [1 - r, 1)$, which is lossless because any certifying σ has $\|\sigma\| \le r$; and equality $U(C) = r$ prunes, which the tight boxes need. One further sound test completes the cascade: D is 1-Lipschitz in the sup norm (for fixed σ every term $\|c_j - v_j\sigma\|$ moves by at most $|c_j - c_j'|$ while the $\|\sigma\|$ term does not move, and minimizing over σ preserves the bound), so a dyadic box C with center c_0 and half-width h satisfies $\max_C D \le D(c_0) + h$ with $D(c_0)$ exactly computable in rational arithmetic; and before the minimax bound $U(C)$ is attempted, the box's center depth is bounded above by the minimum of $G(c_0, \cdot)$ over a fixed float grid on $[0, r] \cup [1 - r, 1)$ (the grid minimum is an upper bound for $D(c_0)$), with a one-in-sixty-four sample of such prunes re-confirmed exactly. The cascade is then: the cheap grid test; the grid-Lipschitz test; the exact Lipschitz box test; the exact minimax test $U(C) \le r$; split.

The method carries its own negative control, and the control behaves exactly as it must. Run on the harmonic instance $v = (1, 2, 3, 4, 5)$ at $r = \delta = \frac{1}{3}$—where the truth is $\rho = 16/47 > \frac{1}{3}$—the branch and bound *refuses to certify*: its survivor set persists around the true deep holes, and the exact depths at the survivors' centers reach $607/1792 \approx 0.3388 > \frac{1}{3}$, at $c = (\frac{11}{256}, \frac{163}{256}, \frac{15}{256}, \frac{79}{256})$. The machinery thus demonstrates both halves of its own soundness profile on one instance: it certifies only what is true, and it localizes the false. The same control distinguishes the two $n = 5$ worlds sharply: for the harmonic instance the surviving excess stays above the threshold by a fixed margin, while for the certified $n = 5$ multiples ($m = 10, 20, 25, 30$) the surviving excess is pure minimax loss and decays with the box size.

The argmax-certified value of Table 2 ($m = 12$) is certified as follows. The exact arrangement analysis is run on the argmax neighborhood (a box of half-width $1/125$ around the located deepest hole): every vertex of the induced hyperplane arrangement inside the box, every ridge candidate, and every box corner is evaluated exactly, and the maximum over the whole box is the displayed value, attained at the recorded point. This certifies that no point of the neighborhood

is deeper than the witness. The two-sided close-out at this instance—pruned leaves outside, center-plus-slack Lipschitz net and exact region analyses over the survivors inside—did not terminate within the stated budgets, and the row stands as an argmax-certified strict refutation with global equality open.

The structural reason the close-out resists is worth recording, because it delimits the method itself. The $m = 12$ argmax is non-dyadic ($5875 = 5^3 \cdot 47$), strictly interior to every dyadic box at every depth, so the box-supremum $U(C)$ exceeds r strictly for every box containing the argmax and the plain pruning rule never fires there; the instances the plain cascade certifies ($m = 16$, 20, 25, 30, and 10) have dyadic (torsion) argmaxes, which is why their trees empty. This is an obstruction to the *plain* cascade, not to the method: a closure at a non-dyadic argmax must let the certificate follow the argmax's rational structure, and the harmonic instance shows that such certificates exist. Two were added. The *sliding-σ gap certificate*: for a fixed integer lift vector m, a point c admits a σ with every runner term $\le r$ iff the interval $[\max_j(c_j - m_j - r)/v_j,\ \min_j(c_j - m_j + r)/v_j]$ intersects $[0, r] \cup [1 - r, 1]$. Write $L(c) = \max_j(c_j - m_j - r)/v_j$ (a maximum of affine functions, hence convex) and $R(c) = \min_j(c_j - m_j + r)/v_j$ (a minimum of affine functions, hence concave); the *gap* is $G(c) = R(c) - L(c)$, a concave function of c, and the certificate holds on the box iff $G \ge 0$ at every point (together with the boundary conditions). A concave function attains its minimum over a box at a vertex, so $G \ge 0$ at all 2^d vertices certifies $G \ge 0$ on the whole box—2^d vertex checks certify the whole box, with σ allowed to *slide* with c along the band, which the single-σ minimax bound cannot do. The *triple-identity certificate* is the weight identity of §5.2 in box form: exhibit three pair-dips of a runner triple with weights $w_{(a,b)} = v_c(v_a + v_b)$ whose weighted values sum, over the box, to a constant $S \le (\sum w)\, r$ (the c-coefficients cancel exactly, verified symbolically per candidate); at every c the minimum dip is then $\le S/\sum w \le r$, and the non-pair terms at each dip are checked $\le r$ by exact affine interval arithmetic. Splitting is lattice-aligned at the dip lattice $q = 470$: dyadic midpoints alone can never align with the balance family's rational slice edges, and boxes straddling them unpruned are exactly what strands the plain cascade. With these, the tree at $r = 16/47$ empties: 782,255 box evaluations (3,721 cheap-grid, 45,598 grid-Lipschitz, 61,191 gap, 197,330 triple-identity, 83,288 exact-minimax prunes; 391,127 splits; maximum depth 81), *zero* survivors, 5,935 s on one core; 4,841 sampled prunes re-confirmed exactly. Combined with either component of the argmax set,

$$\rho(1, 2, 3, 4, 5) \;=\; \tfrac{16}{47} \quad \text{exactly.}$$

The soundness profile matches the rest of the method: adversarial boxes around known deeper holes refuse the new certificates at their thresholds, and the $n = 4$ and $m = 10$ certificates are untouched.

The harmonic sextuple $v = (1, 2, 3, 4, 5, 6)$ was run at its witness level 29/80 and its tree does not empty within the same budget class; its row in Table 2 therefore remains a strict witness, with equality open, and the archive records the asymmetry: at $d = 5$ the survivor regions are larger and more numerous, and the exact region analysis at the harmonic sextuple's argmax does not terminate within the stated budget.

5.5 Summary of the archive

The archive's content, stated as a single paragraph: the covering-radius strengthening of the Lonely Runner statement is false from $n = 5$, measured exactly at the extremal instance ($\rho(1, 2, 3, 4, 5) = 16/47$, certified from both sides) and falsified at four further instances with exact witnesses (29/80 and 4/11; 17/47 argmax-certified and 1793/5000 at the two former boundary rows, both upgraded, the superseded search records $97/288 \to 607/1792 \to 4183/12288$ retained as provenance), boundary rows at $n = 5$ ($m = 6, 7, 8$) at exactly the threshold, and a modular-law extinction record at $n = 6$; the denominators of the exact values are forced by the pair-character identity (Lemma 5.1, Corollary 5.2), whose two archetypes—triple binding and pair bisection—govern every exact value in the archive and explain, and delimit, the recurring 47;

the equivalent center-depth form survives untouched (it is LRC); the deepest-hole modular law is exact at $n = 3$, certified at $n = 4$ (both sides, all tested m, including $m = 16$ by the tree method), false at $n = 5$ (the $k = 1, 3$ multiples fail exactly, $k = 1$ with the exact value $\rho = 16/47 \neq \delta$; $k \in \{2, 4, 5, 6\}$ are certified, each with $\rho = \delta$ exactly), and extinct at $n = 6$; and every number in the paragraph is re-asserted by persisted, independent, exact computation. The archive is finite, small, and open to inspection; that is the point.

6 Conclusion

6.1 What is established

Four facts, each carrying its own verification record. First, the covering-radius form of the Lonely Runner statement—the form that would have opened the geometry-of-numbers toolkit—is strictly stronger than the conjecture and false from $n = 5$, measured exactly at the extremal instance ($\rho = 16/47$ at $v = (1, 2, 3, 4, 5)$, certified from both sides), with the argmax-certified $\rho \geq 17/47$ at $v = (1, 2, 3, 4, 5, 12)$, strict exact witnesses at three further $n = 6$ instances, and an extinction record at $n = 6$; the denominators of the exact values are forced by the pair-character weight identity (Lemma 5.1), which converts the deepest hole at a triple-binding point into a one-line integer computation and delimits the recurring 47. Second, the transfer-matrix route into the strand census is closed by theorem: the lossless cascade's constrained dynamics has no interior rate at any scale, and the survival fraction is exactly 1 at the reflection-symmetric families. Third, the character-rigidity route's natural strengthenings are false with exhaustive exact witnesses, and its marginal envelopes are saturated exactly where the missing margin would have to live. Fourth, on the classification side, the full proof inventory of the program's covering reformulation contains no proved statement of the margin type—verified statement-by-statement—and the one instrument shaped to produce that type is the one the second fact closes.

Against these stands the bounded positive core: the sum-only equality theorem and its lossless finite reformulation; the rigid-zone translate-class classifications at the primes 7, 11, 13, 17, 19, 37; the composite-side structure at p^2 and pq; the conditional scaling theorem with its disclosed hypothesis ledger; and an independently certified deepest-hole archive (equality $\rho = \delta$ at $n = 4$, $m = 16$ and at $n = 5$, $m = 10, 20, 25, 30$, the last closed with an exact Lipschitz box certificate; the exact value $\rho = 16/47$ at the harmonic refutation instance; the argmax-certified $17/47$ at $m = 12$ with global equality open; and strict witnesses at $m = 6, 7, 9$). And beneath both, the mechanism that ties the paper together: losslessness preserves hardness exactly, toolkit strengthenings are false where they matter, and the instruments a faithful reformulation supports compose only into closures. The four instances are four confirmations of the same three-part statement—three on the pair-sum cascade, one independent on the zonotope—and, read together, one thread's worth of evidence for a structural law that should recur in any comparable reformulation culture; the pair-character identity is this thread's own positive piece, and the certified archive is its computational counterpart.

6.2 Why the reformulations cannot cross

It is worth saying plainly what the mechanism predicts, since the prediction is falsifiable and was falsifiable at each step. Any future attack on LRC through a lossless reformulation will face the same fork. On one branch, the attack respects the equivalence and uses only structure invariant under the rewriting; then it is an attack on LRC itself, with the reformulation as notation, and its difficulty is the conjecture's own. On the other branch, the attack strengthens the reformulation—asks for a margin, a gap, a rate, a monotone law— and the strengthening is a strictly stronger statement than LRC; at the extremal instances it is false, and the falsity is exhibited by small exact witnesses. The fork is not a metaphor: in each of the four instances

the program went down one branch, hit exactly the predicted wall, and recorded the wall's coordinates.

The prediction for the zonotope's residual positive question is the same prediction with the sign of interest reversed. Real-rootedness of the h^*-polynomials of the LR-zonotopes—verified through $n = 10$ in both core families and in every sweep instance tested—may well be a true theorem of the arithmetic-matroid type, and if it is, it belongs to the arithmetic-matroid literature, decoupled from the Lonely Runner problem (SI 7). What it cannot be, given the decay and non-correlation records of SI 7 and the falsity of the covering-radius statement it was meant to feed, is a bridge to LRC. That was always the fork's second branch.

6.3 What would be needed

The program's records end with a scope statement rather than a speculation, and we keep to it. A genuine advance on LRC through any of the channels this paper surveys would require an instrument whose output type is a uniform margin over a trivial baseline, independent of the scale parameter, at the extremal instances where every natural candidate saturates or vanishes. No such instrument appears in the covering toolkit, the spectral toolkit, the character toolkit, or the Ehrhart toolkit, and the four closures explain structurally why the obvious candidates from each cannot be one. Whether such an instrument exists anywhere—or whether the conjecture's resistance is exactly this fork, everywhere—is a question this paper does not answer and does not need to answer to justify its archive.

6.4 Scope, stated once more

This paper proves no new cases of the Lonely Runner Conjecture. It does not reduce, weaken, or reformulate-away the conjecture; the surviving equivalent forms are as hard as the original, and the paper's equivalence results make them no easier. The falsified strengthenings were, by construction, stronger than the conjecture, so their falsity leaves the conjecture exactly as it was: open, tight at the harmonic instances, and now surrounded on four sides by precisely mapped walls. The walls are the contribution. They are finite, exact, and checkable, and they are documented so that the next program does not spend its years rediscovering them.

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Supplementary Information:

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Contents

7 The h^*-observations through $n = 10$

This section records the Ehrhart observations on the LR-zonotopes, extended here to $n = 9$ and $n = 10$ in both core families. The extension has a narrow purpose: establish whether the real-rootedness and log-concavity observed at $n \le 8$ continue, and whether the strictness decay and the non-correlation with covering margins persist. Both answers are yes. The section closes with the statement we regard as the residual question of independent interest.

7.1 The table

Table 4 lists, for each family and n: the h^*-polynomial, real-rootedness (verified *exactly* by Sturm-sequence sign-change counts on the square-free part of the integer polynomial, with the 45-decimal-place numerical verification and the companion-matrix check agreeing), log-concavity and Newton's inequalities (exact rational arithmetic), the minimum ratio $h_k^2/(h_{k-1}h_{k+1})$ (the *strictness* S of log-concavity), the Newton floor at the minimizing k, namely $\big((k{+}1)(d{-}k{+}1)\big)/\big(k(d{-}k)\big)$, and the excess ratio S/floor. Every row's Ehrhart data was verified three ways (Eulerian expansion, standard inversion, and an independent depth-first coset count at $t = 1, 2$), the volume identity $e_{n-1} = \sum v_j$ was asserted exactly, and the loneliness identities $\lambda = 1/(n{+}1)$ (harmonic) and $\lambda = 2/(2n{+}1)$ (rung-2) were re-asserted exactly. The $n \le 8$ rows reproduce the program's records digit for digit; the $n = 9, 10$ rows extend them.

Table 4: h^*-polynomials of the LR-zonotopes, $n = 3..10$, both core families. "RR" = real-rooted (exact Sturm verification); "LC/Nw" = log-concave and Newton (exact rational arithmetic); $S = \min_k h_k^2/(h_{k-1}h_{k+1})$; $nS = n \cdot S$; "floor" = the Newton bound $\big((k{+}1)(d{-}k{+}1)\big)/\big(k(d{-}k)\big)$ at the minimizing k (any real-rooted family has $S \ge$ floor there); $S/f = S/\text{floor}$. The $n \le 8$ rows reproduce the program's records digit for digit; the $n = 9, 10$ rows extend them.

family	n	$h^*(z)$	RR	LC/Nw	S	nS	floor	S/f
harm.	3	$1 + 7z + 4z^2$	y	y	12.250	36.8	4	3.06
rung2	3	$1 + 11z + 6z^2$	y	y	20.167	60.5	4	5.04
harm.	4	$1 + 18z + 35z^2 + 6z^3$	y	y	9.257	37.0	3	3.09
rung2	4	$1 + 22z + 51z^2 + 10z^3$	y	y	9.490	38.0	3	3.16
harm.	5	$1 + 37z + 179z^2 + 133z^3 + 10z^4$	y	y	6.511	32.6	9/4	2.89
rung2	5	$1 + 45z + 239z^2 + 181z^3 + 14z^4$	y	y	7.013	35.1	9/4	3.12
harm.	6	$1 + 78z + 744z^2 + 1286z^3 + 399z^4 + 12z^5$	y	y	5.518	33.1	2	2.76
rung2	6	$1 + 86z + 920z^2 + 1682z^3 + 535z^4 + 16z^5$	y	y	5.748	34.5	2	2.87
harm.	7	$1 + 147z + 2522z^2 + 8948z^3 + 7323z^4 + 1201z^5 + 18z^6$	y	y	4.335	30.3	16/9	2.44
rung2	7	$1 + 161z + 3024z^2 + 11144z^3 + 9295z^4 + 1551z^5 + 24z^6$	y	y	4.418	30.9	16/9	2.48
harm.	8	$1 + 290z + 8317z^2 + 51458z^3 + 82947z^4 + 35270z^5 + 3135z^6 + 22z^7$	y	y	3.791	30.3	5/3	2.27
rung2	8	$1 + 298z + 9277z^2 + 60986z^3 + 102275z^4 + 44798z^5 + 4095z^6 + 30z^7$	y	y	3.829	30.6	5/3	2.30
harm.	9	$1 + 559z + 25519z^2 + 258901z^3 + 734059z^4 + 631993z^5 + 155305z^6 + 8035z^7 + 28z^8$	y	y	3.293	29.6	25/16	2.11
rung2	9	$1 + 585z + 28885z^2 + 305035z^3 + 880149z^4 + 764143z^5 + 188659z^6 + 9789z^7 + 34z^8$	y	y	3.323	29.9	25/16	2.13
harm.	10	$1 + 1098z + 77550z^2 + 1213626z^3 + 5529648z^4 + 8309694z^5 + 4180482z^6 + 626910z^7 + 19359z^8 + 32z^9$	y	y	2.987	29.9	3/2	1.99
rung2	10	$1 + 1114z + 83790z^2 + 1377794z^3 + 6457600z^4 + 9866694z^5 + 5017826z^6 + 758710z^7 + 23631z^8 + 40z^9$	y	y	3.004	30.0	3/2	2.00

The sweep families behave identically: every instance of $(1, 2, m)$, $m \le 14$, and $(1, 2, 3, m)$, $m \le 12$, tested by the program was real-rooted, log-concave, and Newton.

7.2 Strictness decay and non-correlation

Two quantitative facts were already in the program's records at $n \leq 8$ and persist through the new rows; both are stated here with the precision the data supports. First, the strictness S decays monotonically in n in both families, with $n \cdot S$ drifting from ≈ 37 at $n = 3, 4$ (harmonic; the rung-2 family starts at 60.5) to ≈ 29.6–30.0 at $n = 9, 10$: within the measured range the decay is well described as $\approx 30/n$. That description cannot persist indefinitely—log-concavity forces $S \geq 1$, and Newton's inequalities force S above the floor $((k{+}1)(d{-}k{+}1))/(k(d{-}k)) \approx 1 + 4/d$ at the middle coefficients, so $nS \approx 30$ would cross the floor near $n \approx 26$—and much of the measured decline is the generic Newton-floor decline that any real-rooted family exhibits. The family-specific content is the excess S/floor, which declines from 3.06 (harmonic) and 5.04 (rung-2) at $n = 3$ to 1.99 and 2.00 at $n = 10$ (Table 4, Figure 1); it, too, declines, so there is no growing quantitative margin in the coefficient structure to spend. Second, the strictness carries no statistically distinguishable correlation with the covering margins at the achievable sample sizes: across the $n = 3$ sweep (twelve instances, $(1, 2, m)$, $m = 3..14$) the Spearman coefficient of strictness against the margin is $+0.32$ ($p = 0.32$); across the $n = 4$ sweep (eight instances, $(1, 2, 3, m)$, $m = 5..12$) it is -0.31 ($p = 0.45$). The sign flips between adjacent rungs and neither coefficient is significant; these arrays are too small to establish correlation or its absence. The harmonic family has margin identically 0 at every n while its strictness varies by a factor of four over the table. Both coefficients were recomputed in the provenance run from the persisted instance data and reproduce exactly.

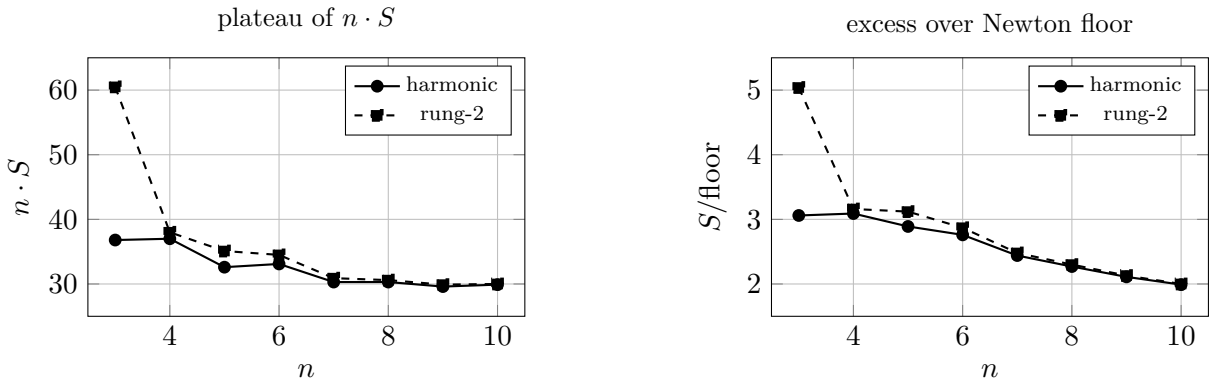


Figure 1: The strictness records of Table 4. Left: the plateau of $n \cdot S$ (the harmonic family starts at 36.8, rung-2 at 60.5; both drift to ≈ 30). Right: the excess S/floor over the Newton floor at the minimizing coefficient—the family-specific content of the decline—which itself decreases from 3.06/5.04 at $n = 3$ to 1.99/2.00 at $n = 10$.

The two facts close the bridge in the coefficient direction, and the falsity of the covering-radius statement (§3.5) closes it in the metric direction: the statement that strict log-concavity would have needed to feed is false at the extremal instances, and the coefficient structure cannot see the metric hole depth that falsifies it. There is no mechanism, and the arithmetic is clean enough that the absence of the mechanism is itself a datum.

7.3 The residual question

What survives is a question of independent arithmetic interest, and we state it decoupled from the Lonely Runner problem entirely. The Ehrhart data of the LR-zonotopes is governed by the gcd structure (2.2) of the projected basis; the h^*-polynomial of the table is the ordinary Ehrhart h^*-polynomial of the lattice zonotope $Z(V)$ (its coefficients carry the gcd structure of the lattice geometry, but the polynomial is the ordinary one—the gcd formula (2.2) is the standard zonotope Ehrhart polynomial). In every instance tested (two core families through $n = 10$, sweeps through

$m \le 14$ at $n = 3$ and $m \le 12$ at $n = 4$), the h^*-polynomial is real-rooted—as it must be, by the theorem below.

Corollary 7.1. *The h^*-polynomial of the LR-zonotope $Z(V)$ of any speed vector $v = (v_1, \dots, v_n)$ with distinct entries and $\gcd(v) = 1$—the Ehrhart h^*-polynomial of the projected-basis zonotope $\pi([0,1]^n) \subset \mathcal{X}_v$ with respect to the lattice $\Lambda = \pi(\mathbb{Z}^n) \cong \mathbb{Z}^{n-1}$—is real-rooted.*

Proof. $Z(V) = \pi([0,1]^n)$ is a lattice zonotope: a projection of the unit cube with vertices $\pi(\{0,1\}^n) \subset \Lambda$, i.e. a zonotope with integer generators $\pi(e_i)$ with respect to $\Lambda \cong \mathbb{Z}^{n-1}$. By Beck–Jochemko–McCullough [14], the h^*-polynomial of every lattice zonotope has only real roots. Hence $h^*(Z(V))$ is real-rooted. □

The real-rootedness is thus a special case of [14], and the data of Table 4 are a confirmation of that theorem on the LR-zonotopes rather than a new observation. What the table records *beyond* [14] is quantitative: the strictness S (the log-concavity margin $h_k^2/(h_{k-1}h_{k+1})$), its decay, and its non-correlation with the covering margins—all properties of the coefficient sequence, not of real-rootedness alone. The data are recorded here and not pursued further: the question belongs to the Ehrhart theory of zonotopes, not to the Lonely Runner problem, and is not a bridge to it.

8 Computational provenance

The reproducibility standard of this paper is one sentence: *every number cited in the text traces to a script that was actually run and whose output was persisted.* The standard is applied uniformly to every cited number, inherited or new. This section is the map.

8.1 The ledger

The provenance run (`scripts/paper_provenance.py`; output `out_paper_provenance.json` and the human-readable `out_paper_provenance.txt`) reads the two persisted run records of the zonotope program and re-asserts, by independent exact computation, every value the paper cites from them: all witness depths (evaluating the runner formulation (2.1) *and* the independent linear-form/lattice-search implementation, asserting exact agreement of the two), all recorded ρ values at their recorded argmax points, all λ and δ values, both Spearman coefficients, both correlation instance arrays (strictness recomputed from the Ehrhart coefficients and margins recomputed from exact λ), and the strictness entries of the $n \le 8$ table. The complete ledger contains 225 items. The number of failures is 0.

The two-way witness verification deserves emphasis because it is the strongest form of check the program has: every witness value in Tables 2 and the fail-witness lists of §5 was recomputed by two implementations with different mathematics (a one-dimensional minimization over the trajectory parameter, and a pruned lattice search in the quotient norm's linear forms), agreeing exactly. A one-line reproduction of the headline witness: evaluate $\min_\sigma \max(\|\sigma\|, \|c_2 - 2\sigma\|, \|c_3 - 3\sigma\|, \|c_4 - 4\sigma\|, \|c_5 - 5\sigma\|)$ at $c = (\frac{46}{47}, \frac{92}{235}, \frac{231}{235}, \frac{35}{47})$; the minimum is attained at $\sigma = \frac{4}{235}$ and the value is $\frac{80}{235} = \frac{16}{47}$.

8.2 The runs behind this paper

The scripts written and run for this paper and its revisions are listed below with their persisted outputs; the inherited records are copied into the paper's artifact bundle.

1. `hstar_n9_n10.py` → `out_hstar_n9_n10.json`: the h^* extension of Appendix 7 ($n = 3..10$, both families; Ehrhart asserted three ways; roots at 45 decimal places on exact integer coefficients; strictness, Newton, log-concavity exact; λ asserted against the classical values).

2. `certify_rung2_n5.py` → `out_certify_rung2.json`: the branch-and-bound certificate of §5.4, including the cross-validation of the two D implementations (forty random rational points per instance, exact agreement), the controls ($n = 3$ and $n = 4$ core instances, all certified), the negative control (harmonic $n = 5$, correctly refusing), and the ladder targets (§5).

3. `paper_provenance.py` → `out_paper_provenance.{json,txt}`: the 225-item ledger above.

4. `certify_followup.py` → `out_certify_followup.json`: the survivor enclosure analysis of the open $m = 10$ run and the recovered improved harmonic witness (607/1792 at $c = (\frac{11}{256}, \frac{163}{256}, \frac{15}{256}, \frac{79}{256})$).

5. `rho_closeout.py` → `out_rho_closeout.json`: the exact-argmax determination of the sharpest witnesses (grid and congruence-family localization of the deepest holes, exact arrangement analysis of the argmax regions, the binding-structure extraction of §5.2 with every binding value, translate, and χ-residual re-computed by both D implementations, and the branch-and-bound runs at the exact levels).

6. `closeout_certify.py` and `lean_closeout.py` → `out_closeout_*.json` and `out_lean_*.json`: the two-sided close-out attempts at the exact levels: the branch-and-bound survivor collection at the exact level (leaves certify $D \le r$ outside the survivor boxes), the center-plus-slack Lipschitz net over the survivors, and the exact arrangement analysis of the surviving regions (§5.4, Table 5).

7. `m9_witness_hunt.py` and `m9_ridge_scan.py` → `out_m9_witness.json`: the strict $m = 9$ witness of Table 2 (multi-start exact hill-climb, then the parametrized binding-family scan that produced the 1793/5000 point and its chain binding structure).

8. `local_refine.py`: the reusable float-grid localizer used to seed the exact arrangement analyses.

9. `certify_m10_v2.py` → `out_certify_m10_v2.json`: the $m = 10$ close-out of §5: the sound multi-slice search screen (4.2 million random points, dyadic grids, torsion neighborhood, climbs, Nelder–Mead polishes, 114 exact re-evaluations) and the re-run branch and bound with the exact Lipschitz box test added to the cascade (190,295 box evaluations, 95,148 certified leaves, zero survivors, 1,286 s; the negative control refusing, the $n = 4$ certificates re-certifying with identical counts, 846/846 sampled prunes re-confirmed exactly).

10. `certify_harmonic_v4.py` → `out_certify_harmonic_v4.json`: the harmonic close-out of §5.4: the balance-family search (the 30,000 survivor seeds of the control run, the family witness $c(35)$, the identity $32\,r_{(3,5)} + 27\,r_{(4,5)} + 35\,r_{(3,4)} = 32$ verified exactly at every member), and the branch and bound at $r = 16/47$ with the sliding-σ gap and triple-identity certificates plus lattice-aligned splitting at the dip lattice $q = 470$ (782,255 box evaluations, 391,128 certified leaves, zero survivors, 5,935 s; 4,841/4,841 sampled prunes re-confirmed exactly); includes the $n = 6$ $m = 9$ and $m = 12$ refutation searches with their exactly evaluated dyadic witnesses.

11. `crosscheck_cert_witnesses.py`: the cross-validation of the two closure runs' witness-level claims against the independent linear-form/lattice-search implementation of this paper's ledger: seven exact evaluations (the $m = 10$ torsion center at 7/22; the balance family at $k = 20, 27, 35$, each exactly 16/47; the $m = 9$ dyadic search point at its exact depth; the $m = 12$ and $m = 9$ clean witnesses at 17/47 and 1793/5000), zero failures, plus the binding-structure extraction at the four $n = 5$-multiple center classes (pair $(1, m)$ bisecting at each), the verification behind the archetype remark of §5.2.

8.3 Certificates and budgets

The branch-and-bound certificates are finite objects: a certificate for $\rho(V) \le r$ is the pruned dyadic tree, each leaf carrying an exact rational U-value $\le r$ computed from the kink/crossover enumeration of §5.4. Budgets and certificate sizes for the runs cited in the text are listed in Table 5; the JSON persists leaf samples, survivor aggregates (count, bounding box, maximum excess over r, exact depths at sampled survivor centers), and the exact U-values, so the recorded leaves and survivor aggregates can be re-audited without re-running the search, and re-running the search reproduces every certificate deterministically (a full leaf-by-leaf re-audit of the largest runs requires the re-run). The two closure runs persist, in addition, their per-certificate fire counts (including the gap and triple-identity counts of the harmonic cascade), the sampled-prune exact re-confirmations, and the negative-control refusals; and their witness-level claims were re-asserted by an independent cross-validation run (`crosscheck_cert_witnesses.py`, seven exact evaluations, zero failures) before incorporation into this paper.

Table 5: Branch-and-bound runs cited in the text. "leaves" = certified boxes (each with exact $U \le r$, or an exact certificate of the added Lipschitz/gap/triple-identity type); "surv." = unresolved boxes at cap or at budget; wall-clock seconds in parentheses. "Direct closure" = running the tree to an empty heap at exactly r with the plain cascade; it is unavailable at non-dyadic argmaxes (§5.4), which the plain-cascade rows record. "Close-out" = the two-sided assembly of leaves, the center-plus-slack Lipschitz net, and the exact region analyses (`lean_closeout.py`); "budget-open" records that the region analyses did not terminate within the stated budget. The two "certified" rows at $r = 16/47$ and at $m = 10$'s $r = 7/22$ are the closures: the enhanced cascade (gap + triple-identity, lattice-aligned splitting) and the Lipschitz box test respectively.

instance	r	verdict	leaves	surv.
$(1,2,3)$	1/4	certified	14 (0.0)	0
$(1,2,6)$	3/14	certified	20 (0.0)	0
$(1,2,3,4)$	3/10	certified	438 (2)	0
$(1,2,3,8)$	5/18	certified	156 (1)	0
$(1,2,3,12)$	7/26	certified	168 (1)	0
$(1,2,3,16)$	9/34	certified	222 (2)	0
$(1,2,3,4,5)$	1/3	*refused* (control)	17,520 (421)	45,355
$(1,2,3,4,5)$	16/47	refused (direct closure; non-dyadic argmax)	49,422 (3002)	57,219
$(1,2,3,4,5)$	16/47	close-out (budget-open; 6,835 net-certified, 8 regions)	42,914 (1400)	52,581
$(1,2,3,4,5)$	16/47	*certified* (enhanced cascade: gap + triple-identity, lattice-aligned splitting)	391,128 (5935)	0
$(1,2,3,4,5,12)$	17/47	refused (direct closure)	33,926 (3004)	15,709
$(1,2,3,4,5,12)$	17/47	close-out (budget-open; 1,270 certified, 14 regions)	30,340 (1400)	9,749
$(1,2,3,4,5,6)$	29/80	refused (direct closure)	52,772 (3002)	42,822
$(1,2,3,4,10)$	7/22	first attempt open (residual volume $< 6 \times 10^{-7}$)	61,038 (1501)	21,153
$(1,2,3,4,10)$	7/22	*certified* (Lipschitz box test; sound search screen)	95,148 (1286)	0
$(1,2,3,4,20)$	13/42	certified	37,018 (981)	0
$(1,2,3,4,25)$	4/13	certified	10,414 (231)	0
$(1,2,3,4,30)$	19/62	certified	10,646 (274)	0

8.4 Reproduction

All scripts run under CPython 3.12 with `fractions`, `numpy`, `scipy`, and `mpmath`, and take minutes to hours on one core; the certificates of Table 5 take seconds to minutes (the longest plain-closure certificate, $m = 20$, took sixteen minutes; the enhanced-cascade closures took twenty-one minutes at $m = 10$ and ninety-nine minutes at the harmonic instance), and the refused control runs were

run to their stated budgets. One-line reproduction from the artifact bundle's root (each writes its `out_*.json` next to itself, deterministically):

- `python3 hstar_n9_n10.py`
- `python3 certify_rung2_n5.py`
- `python3 paper_provenance.py`
- `python3 certify_followup.py`
- `python3 rho_closeout.py` (phases: `grid47`, `tiny_exact`, `structure`, `certify`)
- `python3 lean_closeout.py 1,2,3,4,5 16/47` (and the $m = 12$, $m = 9$, harmonic-$n = 6$ analogues)
- `python3 certify_m10_v2.py` and `python3 certify_harmonic_v4.py` (the two closure runs)
- `python3 crosscheck_cert_witnesses.py` (the seven-witness cross-validation)

The artifact bundle accompanying the paper contains the scripts, their outputs, the inherited zonotope records (`out_lrc_zono_final.json`, `out_lrc_zono_rescan.json`, and the library `lrc_zono_lib.py` they import), and the ledger. Nothing in the paper relies on an unlogged computation, a hard-coded constant, or a claim of the form "verified elsewhere".

9 A self-contained verification core

This section verifies the sharpest negative claims self-containedly, without the companion monograph. It collects the elementary identity behind the four forms, the runner formulation with its certificate, the explicit statement of the sum-only equality, and one-line hand checks of the headline witnesses.

9.1 The identity and the runner formulation

Lemma 9.1 (depth identity)**.** *For every $t \in \mathbb{T}$, $\max_{1\le j\le n}\|\frac{1}{2} - v_j t\| = \frac{1}{2} - \min_{1\le j\le n}\|v_j t\|$. Consequently $\delta(V) = \frac{1}{2} - \lambda(V)$ with $\delta(V) = \mathrm{dist}_\infty\big(\mathbb{R}v,\ \mathbb{Z}^n + \frac{1}{2}\mathbf{1}\big)$, and the δ-form of §2 is exactly* LRC_n.

Proof. Each map $x \mapsto \|x\|$ satisfies $\|\frac{1}{2} - x\| = \frac{1}{2} - \|x\|$: reflecting about $\frac{1}{2}$ exchanges the two half-intervals and fixes the midpoint. Taking $x = v_j t$ and maximizing over j gives the displayed identity (a maximum of quantities each equal to $\frac{1}{2}$ minus the corresponding term is $\frac{1}{2}$ minus the minimum). The distance form is the same statement along the trajectory line tv, $t \in \mathbb{R}$. □

The computations of this paper use the *runner formulation* (2.1), restated here:

$$D(c) \;=\; \min_{\sigma\in[0,1)}\ \max\Big(\|\sigma\|,\ \max_{2\le j\le n}\|c_j - v_j\sigma\|\Big).$$

Two facts make it auditable. First, it is the quotient-norm depth $\mathrm{dist}_{\mathcal{X}_v}(c, \Lambda)$ written in the c-coordinates of §2: the linear forms ϕ^{1j}, ϕ^{ij} listed there generate the norm, the parameter σ is the free coordinate x_1 of a lift, and minimizing over σ minimizes over the lift. Second, it is *Lipschitz-auditable*: each term $\|c_j - v_j\sigma\|$ is v_j-Lipschitz in σ and the whole objective is piecewise linear with breakpoints at the half-integers $c_j - \frac{1}{2} - v_j\sigma \in \mathbb{Z}$, so the minimum is attained at a rational breakpoint or crossing and is exactly computable by hand or by exact rational arithmetic. The independent linear-form/lattice-search implementation of the program agrees with it exactly on every instance used in this paper (forty random rational points per instance).

9.2 The sum-only equality, explicitly

Theorem 9.2 (sum-only equality; explicit form)**.** *Let $V = (v_1, \dots, v_n)$ have distinct positive entries with $\gcd(V) = 1$ and $0 < M(V) < \frac{1}{2}$. For each unordered pair $\{u, w\} \subset [n]$ write $N_{uw} = v_u + v_w$ and*

$$\tau(u, w) \;=\; \max_{0 \le k < N_{uw}} \; \min_{p \in [n] \setminus \{w\}} \; \| \frac{v_p k}{N_{uw}} \|.$$

Then $M(V) = \max_{\{u,w\}} \tau(u, w)$, and consequently LRC_n holds for all V if and only if for every V there is a pair $\{u, w\}$ with $\tau(u, w) \ge 1/(n+1)$.

The two ingredients (proved in the companion monograph, Theorem 1.1 there; the finitization is what every instance of §3.2 entered through) are: (i) at a global optimum t^* with $0 < M(V) < \frac{1}{2}$ the binding runners can be chosen two-sided—one rising, one falling—which forces $(v_u + v_w)t^* \in \mathbb{Z}$ for that pair, so t^* lies on the *sum* lattice of denominator $v_u + v_w$; and (ii) on that lattice the two binding runners are exact reflections ($\|v_u k/N\| = \|v_w k/N\|$ for all k, since $v_u \equiv -v_w \pmod N$), so the n-runner minimum equals an $(n-1)$-runner minimum. The companion's Lemma 2.1 also records the boundary cases $M(V) = \frac{1}{2}$ (all speeds odd) and $M(V) = 0$ (impossible).

9.3 Hand checks of the headline witnesses

All the strict witnesses of Table 2 are one-minute hand computations with (2.1); the two closed values need the branch-and-bound half of their certificates as well (§5.4), but the witness halves are hand checks too.

The harmonic argmax value $16/47$. Take $v = (1, 2, 3, 4, 5)$ and $c = (\frac{122}{1175}, \frac{688}{1175}, \frac{284}{1175}, \frac{80}{1175})$. At $\sigma = \frac{96}{1175}$ (working in 1175-ths, with $c = (\frac{122}{1175}, \frac{688}{1175}, \frac{284}{1175}, \frac{80}{1175})$): $\|\sigma\| = \frac{96}{1175}$; $\|c_2 - 2\sigma\| = \|\frac{122-192}{1175}\| = \frac{70}{1175}$; $\|c_3 - 3\sigma\| = \|\frac{688-288}{1175}\| = \frac{400}{1175}$; $\|c_4 - 4\sigma\| = \|\frac{284-384}{1175}\| = \frac{100}{1175}$; $\|c_5 - 5\sigma\| = \|\frac{80-480}{1175}\| = \frac{400}{1175}$. The maximum is $\frac{400}{1175} = \frac{16}{47}$, attained twice (runners 3 and 5, slopes $+3$ and -5 opposite), and it is minimized: any local shift of σ increases one of the two. Hence $D(c) = 16/47 > \frac{1}{3} = \mathrm{thr}_5$. The pair-form side of the same point is the quadruple binding of §5.2: $\phi^{45}(c - z) = -16/47$ at $z = (0, 0, 0, -1)$, $\phi^{34}(c - z) = -16/47$ at $z = (0, 1, 0, 0)$, $\phi^{35}(c - z) = +16/47$ at $z = 0$, and $\phi^{25}(c - z) = +16/47$ at $z = (0, 1, 1, 1)$, each a two-term integer evaluation.

The rung-2 argmax value $17/47$. Take $v = (1, 2, 3, 4, 5, 12)$ and $c = (\frac{4714}{5875}, \frac{1}{8}, \frac{5}{8}, \frac{6}{25}, \frac{2034}{5875})$. At $\sigma = \frac{482}{5875}$ the six terms are $\frac{482}{5875}$, $\frac{2125}{5875}$, $\frac{5693}{47000}$, $\frac{13951}{47000}$, $\frac{1000}{5875}$, $\frac{2125}{5875}$: the maximum $\frac{2125}{5875} = \frac{17}{47}$ is attained twice (runners 2 and 12, slopes $+2$ and -12 opposite) and is minimized, so $D(c) = 17/47 > \frac{5}{14} = \mathrm{thr}_6$. The binding triple $\{2, 5, 12\}$ and its translates are recorded in §5.2.

The strict witnesses. For $v = (1, 2, 3, 4, 5, 6)$ take $c = (0, \frac{5}{16}, \frac{15}{16}, \frac{3}{4}, \frac{1}{2})$ and $\sigma = \frac{23}{160}$: the six terms are $\frac{23}{160}$, $\frac{23}{80}$, $\frac{19}{160}$, $\frac{29}{80}$, $\frac{1}{32}$, $\frac{29}{80}$, so $D(c) = 29/80 > \frac{5}{14}$ (runners 4 and 6 tie at the maximum with opposite slopes). For $v = (1, 2, 3, 4, 5, 7)$ take $c = (\frac{1}{16}, \frac{9}{16}, 0, \frac{3}{8}, 0)$ and $\sigma = \frac{1}{11}$: the six terms are $\frac{1}{11}$, $\frac{21}{176}$, $\frac{51}{176}$, $\frac{4}{11}$, $\frac{7}{88}$, $\frac{4}{11}$, so $D(c) = 4/11 > \frac{5}{14}$. For $v = (1, 2, 3, 4, 5, 9)$ take $c = (\frac{1}{32}, \frac{15}{32}, \frac{1383}{10000}, \frac{32921}{40000}, \frac{95473}{200000})$ and $\sigma = \frac{18577}{200000}$: the six terms are $\frac{18577}{200000}$, $\frac{3863}{25000}$, $\frac{38019}{200000}$, $\frac{5831}{25000}$, $\frac{1793}{5000}$, $\frac{1793}{5000}$, so $D(c) = 1793/5000 > \frac{5}{14}$ (runners 5 and 9 tie at the maximum).

The $7/22$ *center class (the* $m = 10$ *ladder closure's witness).* Take $v = (1, 2, 3, 4, 10)$ and $c = (\frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2})$. At $\sigma = \frac{7}{22}$ the five terms of (2.1) are $\frac{7}{22}$, $\frac{3}{22}$, $\frac{1}{22}$, $\frac{5}{22}$, $\frac{7}{22}$: explicitly $\|\sigma\| = \frac{7}{22}$; $\|\frac{1}{2} - 2\sigma\| = \frac{3}{22}$; $\|-3\sigma\| = \frac{1}{22}$; $\|\frac{1}{2} - 4\sigma\| = \frac{5}{22}$; and $\frac{1}{2} - 10\sigma = -\frac{59}{22} \equiv \frac{7}{22} \pmod 1$, so the fifth term is $\frac{7}{22}$. The maximum is $\frac{7}{22}$, and it is minimized: the binding terms are the σ-term (slope $+1$) and the fifth term (slope -10), opposite signs, so any local shift of σ increases one of them. Hence $D(c) = 7/22$, and the branch-and-bound certificate of §5.4 ($\rho \le 7/22$, the tree emptying) upgrades this to $\rho(1, 2, 3, 4, 10) = \frac{7}{22}$ exactly; since $\delta = 1/2 - \lambda = 1/2 - 2/11 = 7/22$, the deepest hole at $m = 10$ is the center class itself. The pair-form side is the bisection of §5.2: the pair $(1, 10)$ binds at two translates with opposite signs, $2 \cdot 11 \cdot \rho = 7 \in \mathbb{Z}$.

The balance-family witness at the harmonic argmax. Take $c = c(35) = (\frac{46}{47}, \frac{92}{235}, \frac{231}{235}, \frac{35}{47})$. At $\sigma = \frac{4}{235}$ the five terms of (2.1) are $\frac{4}{235}$, $\frac{13}{235}$, $\frac{80}{235}$, $\frac{20}{235}$, $\frac{80}{235}$: explicitly $\|\sigma\| = \frac{4}{235}$; $\|\frac{46}{47} - 2\sigma\| =$

$\|\frac{222}{235}\| = \frac{13}{235}$; $\|\frac{92}{235} - 3\sigma\| = \frac{80}{235}$; $\|\frac{231}{235} - 4\sigma\| = \|\frac{215}{235}\| = \frac{20}{235}$; and $\|\frac{35}{47} - 5\sigma\| = \|\frac{155}{235}\| = \frac{80}{235}$. The maximum is $\frac{80}{235} = \frac{16}{47}$, attained with the third and fifth terms binding (slopes -3 and $+5$, opposite), and the tie over all three dips of the triple $\{3, 4, 5\}$—forced by the identity along the family—gives the same value at every member; the two-sided certificate of §5.4 then makes 16/47 the exact covering radius.

Every value in Table 2 was additionally re-asserted by two independent implementations in the provenance run (Appendix 8), and the closure runs' witnesses by the cross-validation run described there.